



Ch3: Chemical Bonding

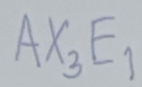
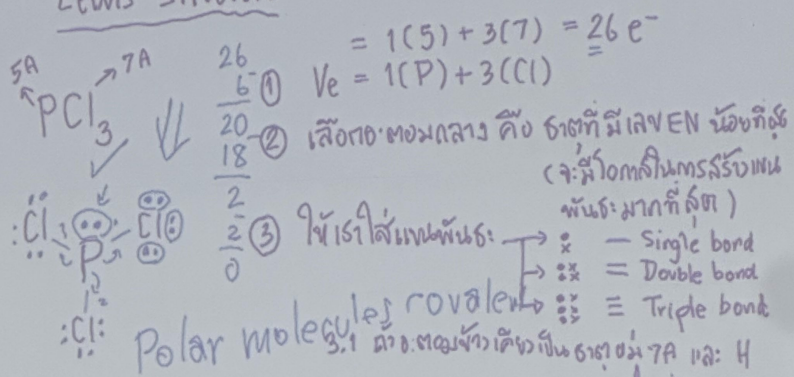
Ref: Ralph H. Petrucci, F. Geoffrey Herring, Jeffry D. Madura, Carey Bissonnette. General Chemistry: Principles and Modern Applications. Prentice Hall 2010, English 10th ed.

Theodore E. Brown, H. Eugene H LeMay, Bruce E. Bursten, Catherine Murphy, Patrick Woodward. Chemistry: The Central Science. Prentice Hall 2011, English 12th ed.

Raymond Chang and Jason Overby. General Chemistry: The Essential Concepts. McGraw-Hill Science/Engineering/Math 2010, 6th ed.

Ask Copilot

Lewis structure

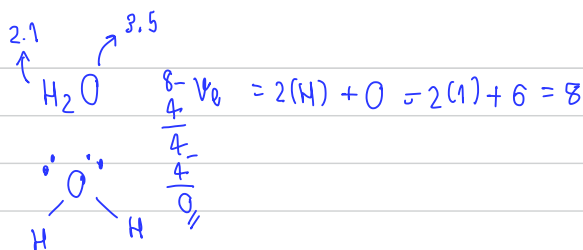


④ ทำให้อิเล็กตรอนของกลางได้ single bond ครบ 8 ตามกฎ Octet

VSEPR

↓ AB_mE_n

AX_mE_n → อิเล็กตรอนคู่โดดเดี่ยวของอ.ของกลาง
 → จำนวน lone pair e⁻ ของอ.ของกลาง
 (-, =, ≡)
 → จำนวนพันธะของอ.ของกลาง



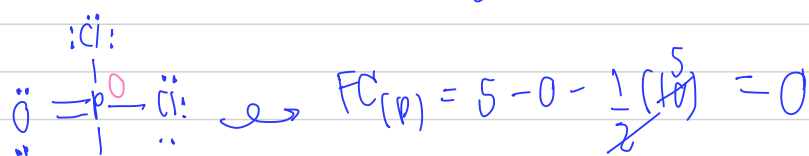
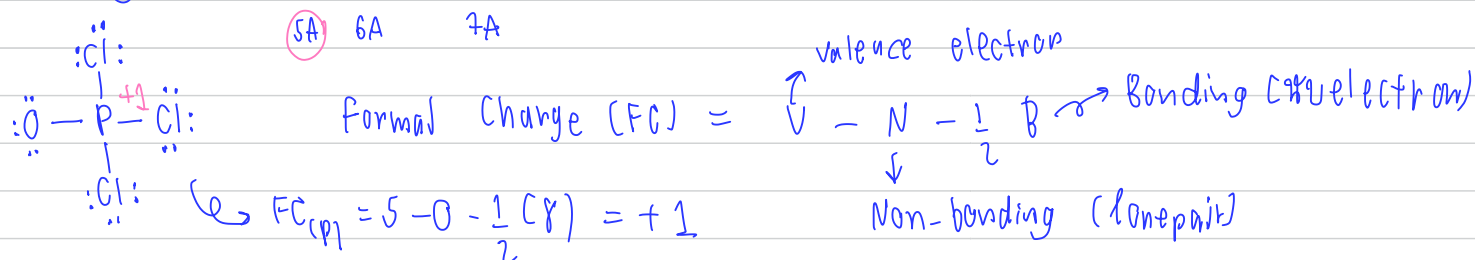
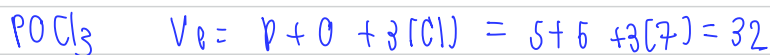
$\text{AX}_2\text{E}_2 \Rightarrow \text{Polar}$

ขนาดของพันธะไอออน $\Rightarrow \Delta \text{EN}$

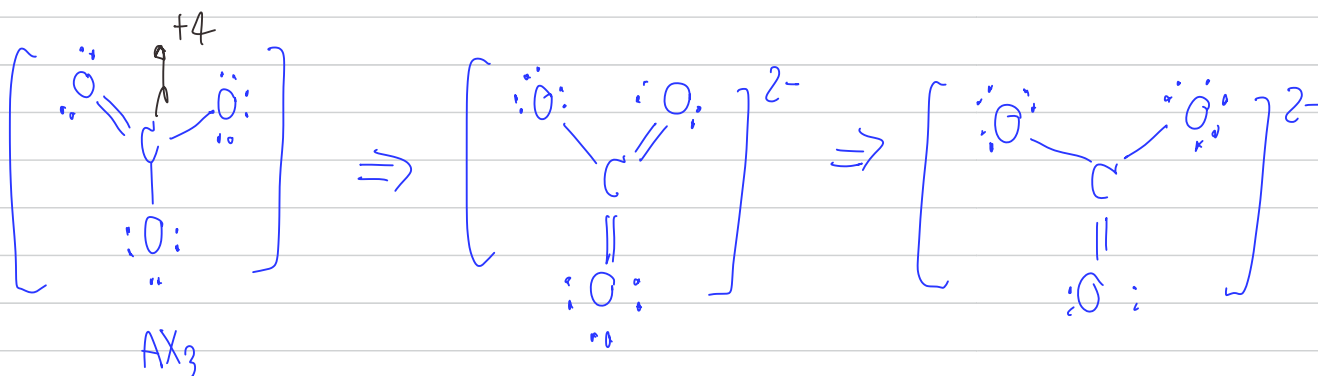
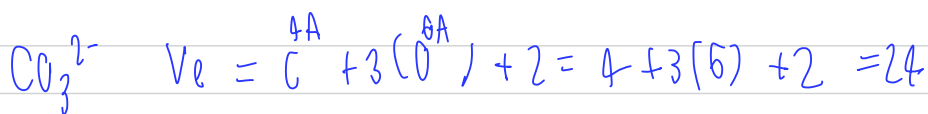
① $\Delta \text{EN} = 0$ Nonpolar Covalent Bond (NPC)

② $0 < \Delta \text{EN} < 1.8$ Polar covalent Bond (PCB)

③ $\Delta \text{EN} \geq 1.8$ Ionic Bond (IB)

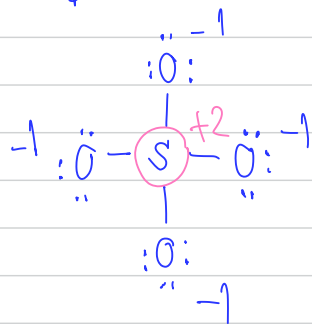


Non polar.



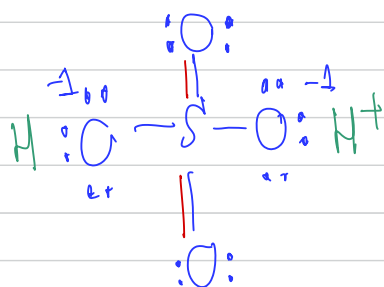
Resonance

$$SO_4^{2-} \quad V_e = 32 = 6 + 4(6) + 2 = 6 + 4(6) + 2$$



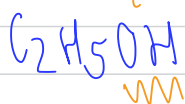
$$\begin{aligned} FC(S) &= V - N - \frac{1}{2}(B) \\ &= 6 - 0 - \frac{1}{2}(8) \\ &= +2 \end{aligned}$$

$$FC(O) = 6 - 6 - \frac{1}{2}(2) = -1$$



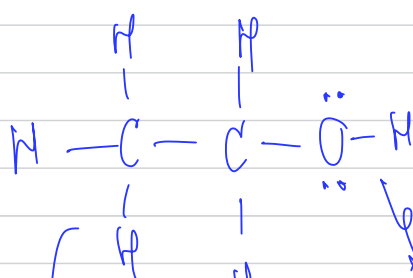
$$\begin{aligned} FC(S) &= V - N - \frac{1}{2}(B) \\ &= 6 - 0 - \frac{1}{2}(12) \\ &= 0 \end{aligned}$$

hydroxyl group H_2VO_4
 $2H^+ + VO_4^{2-}$



$$V_e = 2(C) + 6(H) + 0$$

$$= 2(4) + 6(1) + 0 = 20 e^-$$



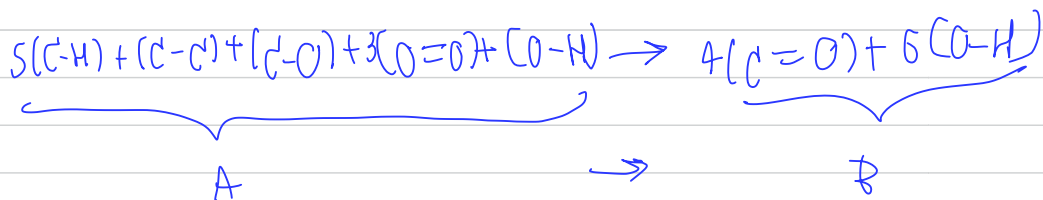
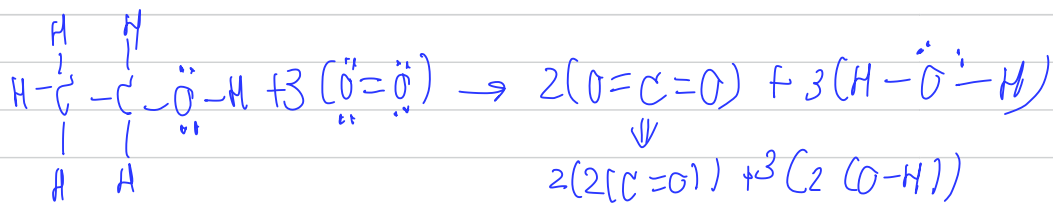
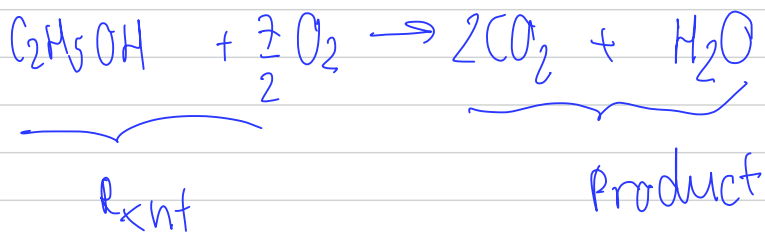
AX_4 $AX_2E_2 \Rightarrow$ Polar

Non polar

$$\Delta H_{rxn} = \sum B.E_{rxnt} - \sum B.E_{product}$$

Heat of reaction (ΔH)

$$\Delta H = \sum (v_i - v_j)$$



exothermic

$$\ominus \Delta H_{rxn} = \sum B.E_{rxn} - \sum B.E_{product}$$

Endothermic

$$= A - B \quad \text{kJ/mol}$$



Topics

- Chemical bonds/Chemical bonding
- Lewis structure/ Molecular structure: The VSEPR model
- Types of chemical bonding (Covalent, Ionic and Metallic Bonds)
- Naming compounds/Bond energy (Heat of reaction/Lattice energy)
- Valence bond theory (VB)
- Molecular orbital theory (MO)



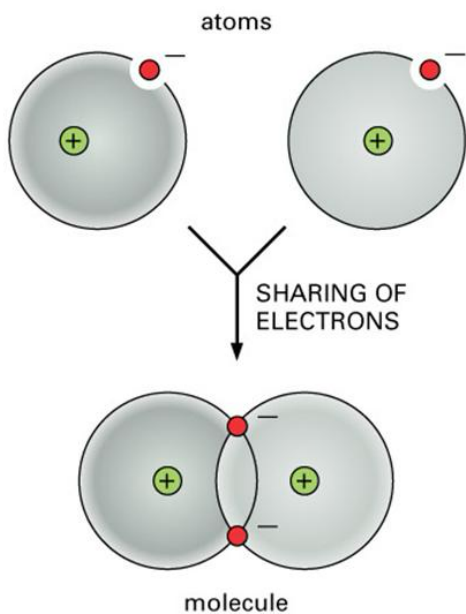
พันธะเคมี (Chemical bond) แรงดึงดูดระหว่างอะตอมที่ทำให้อะตอมอยู่ด้วยกันในรูปของโมเลกุล

แรงยึดเหนี่ยวภายในโมเลกุล

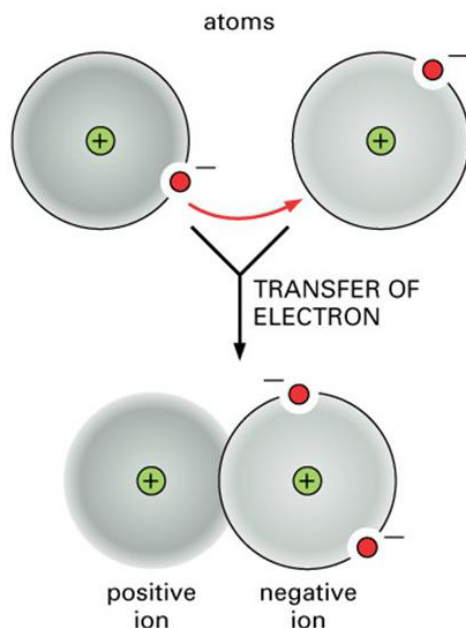
พันธะโคเวเลนต์

พันธะไอออนิก

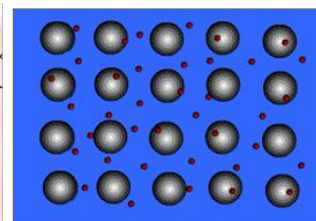
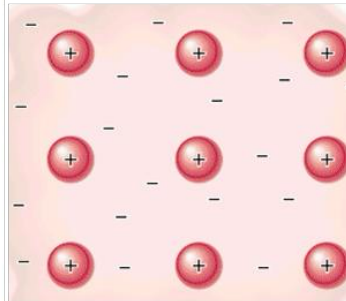
พันธะโลหะ



covalent bond



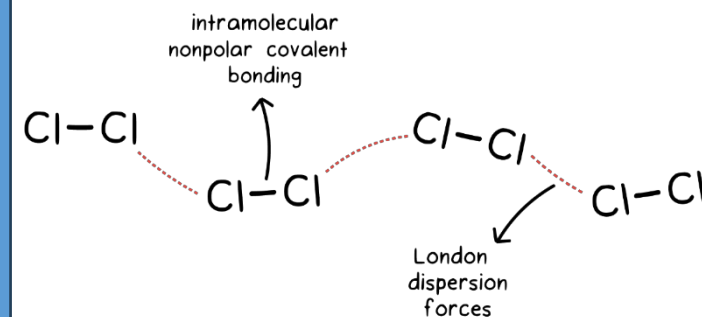
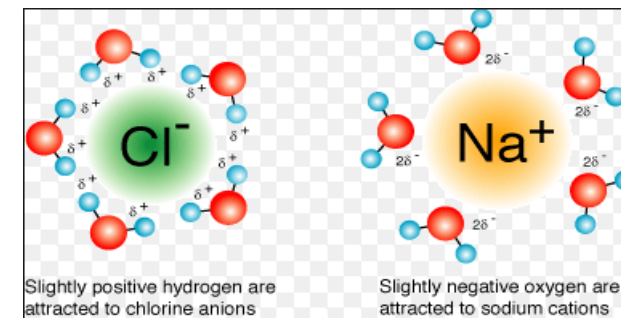
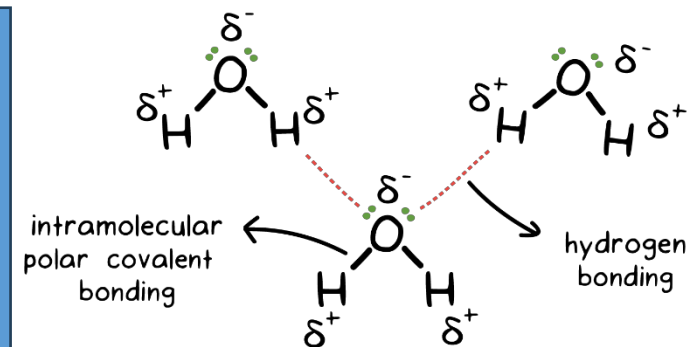
ionic bond



แบบจำลองทะเลอิเล็กตรอน (electron sea model)
เวเลนซ์อิเล็กตรอนเคลื่อนที่อย่างอิสระ แรงดึงดูดที่อิเล็กตรอนมีต่อไอออนบวกมากกว่าแรงผลักระหว่างอิเล็กตรอนด้วยกันเอง ทำให้โลหะมีโครงสร้างผลึกเสถียร

แรงยึดเหนี่ยวระหว่างโมเลกุล :

- แรงลอนดอน
- พันธะไฮโดรเจน
- แรงดึงดูดระหว่างขั้ว



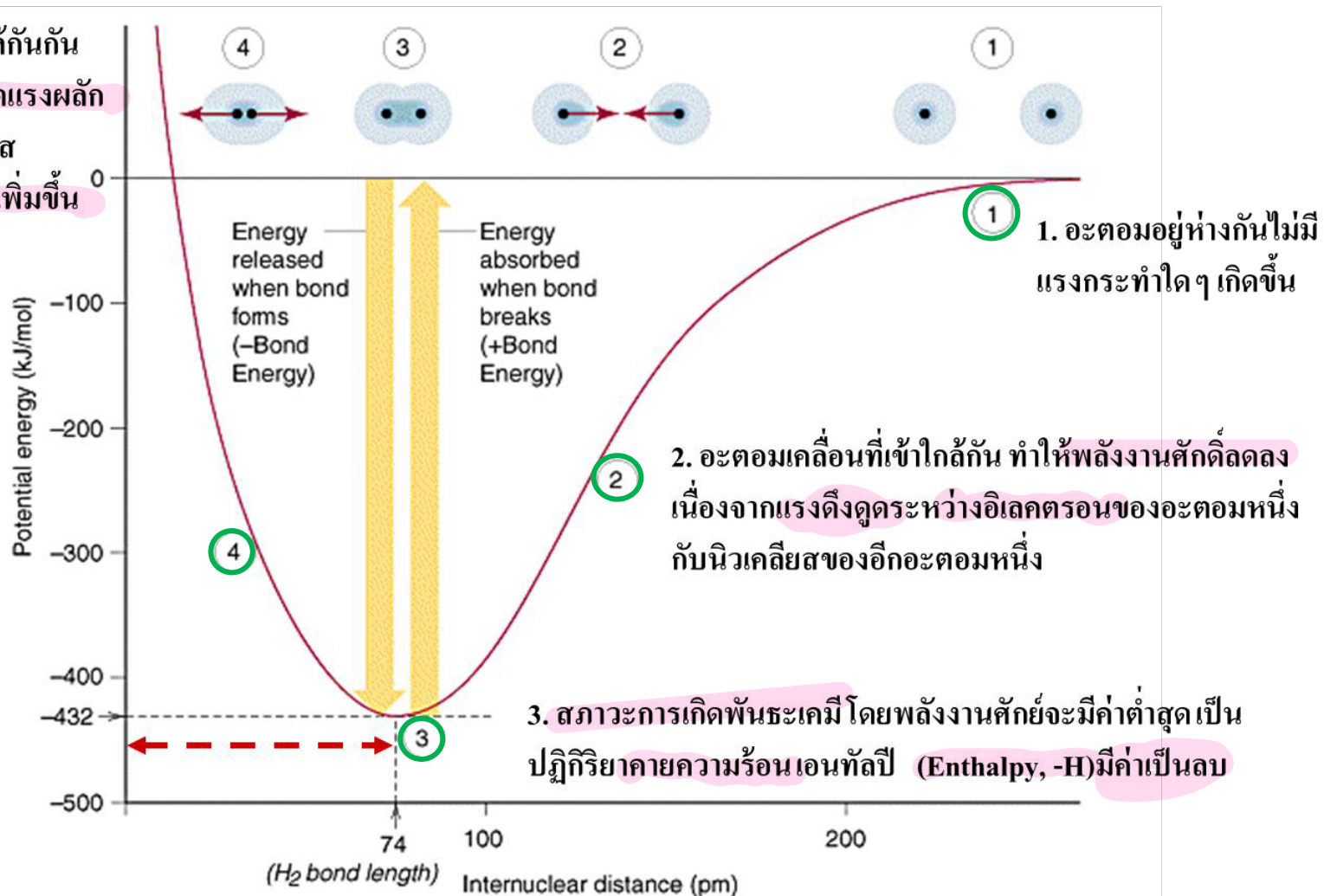
เกิดจากการใช้อิเล็กตรอนร่วมกัน
ระหว่างอะตอมที่จะเกิดพันธะ

เกิดจากการโยกย้ายของอิเล็กตรอน
จากอะตอมหนึ่งไปยังอีกอะตอมหนึ่ง

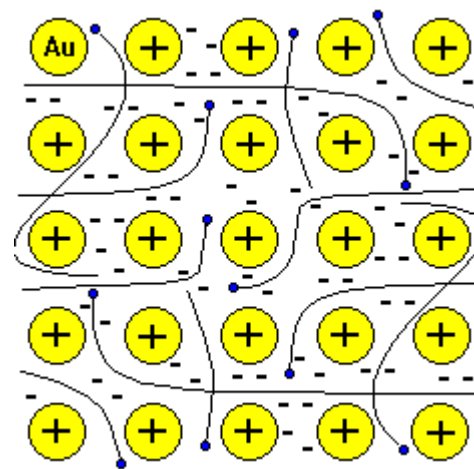
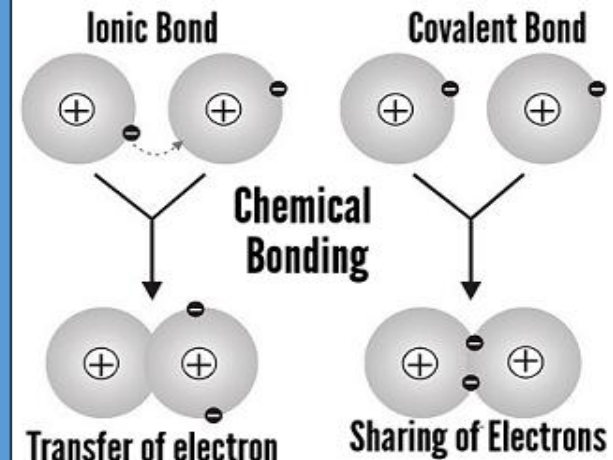


การเกิดพันธะเคมี (chemical bonding formation)

4. อะตอมเข้าใกล้กันมากขึ้น ทำให้เกิดแรงผลักระหว่างนิวเคลียส พลังงานศักย์จึงเพิ่มขึ้น



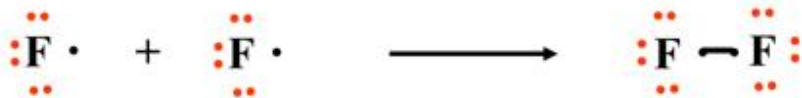
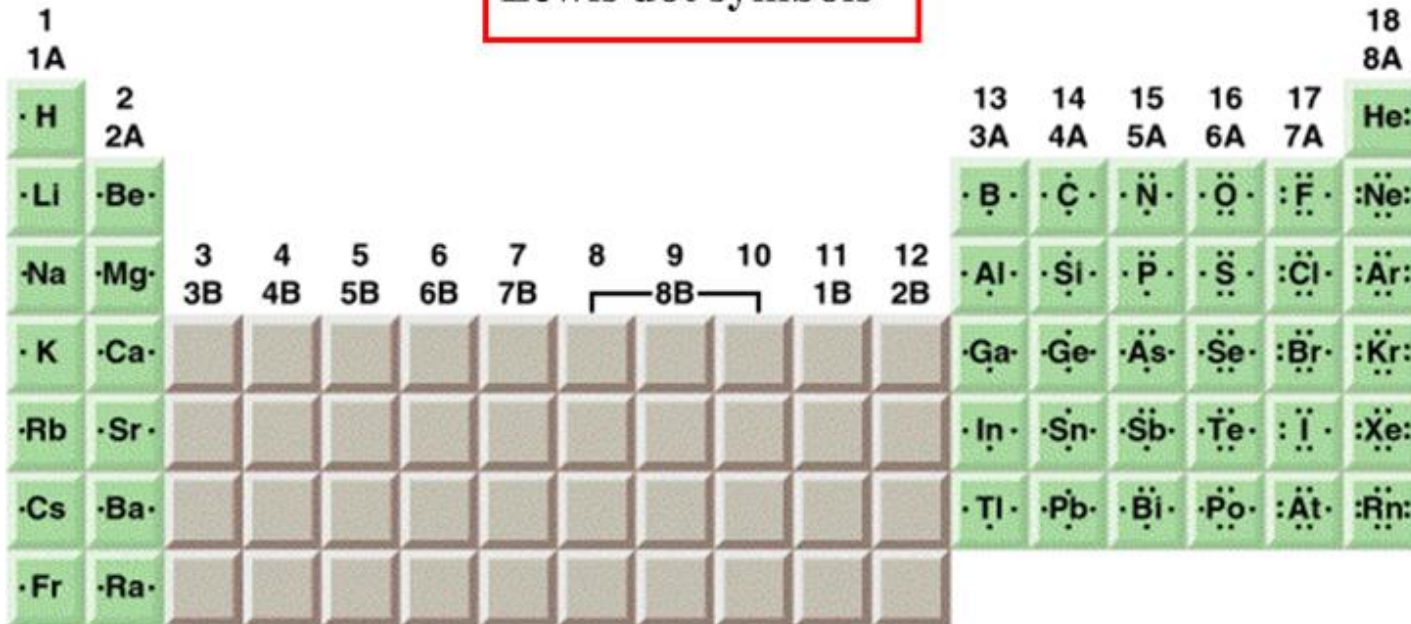
ความยาวพันธะ คือระยะห่างระหว่างอะตอมที่มีพลังงานศักย์ต่ำที่สุด (เสถียรมากที่สุด)





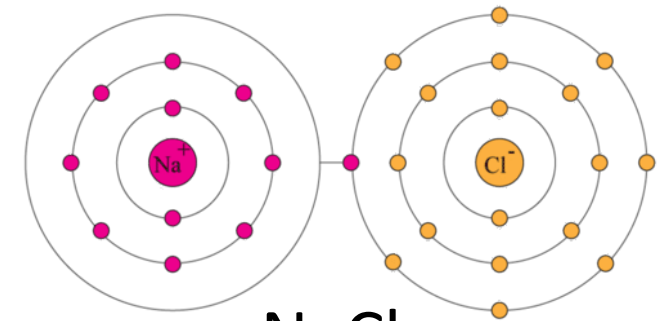
Lewis Structure

Lewis structures (also known as Lewis dot diagrams, Lewis dot formulas, Lewis dot structures, and electron dot structures) are diagrams that show the bonding between atoms of a molecule and the lone pairs of electrons that may exist in the molecule



Ex. Covalent Bond

หมู่ คาบ	1A	2A	3A	4A	5A	6A	7A	8A
	ns^1	ns^2	ns^2np^1	ns^2np^2	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6
2	• Li	• Be •	• B •	• C •	• N •	• O •	• F •	• Ne •
3	• Na	• Mg •	• Al •	• Si •	• P •	• S •	• Cl •	• Ar •



NaCl

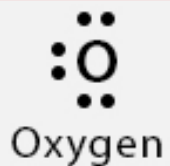
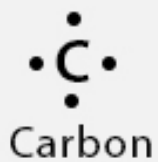
Ex. Ionic Bond



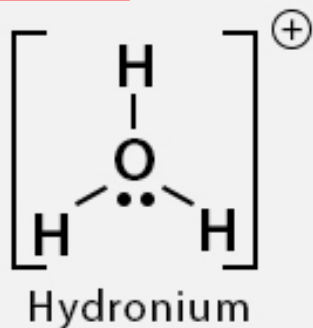
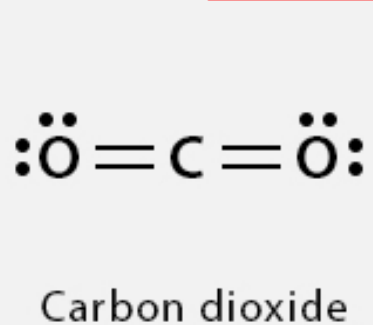
Lewis Structure

Lewis Structure

Atoms



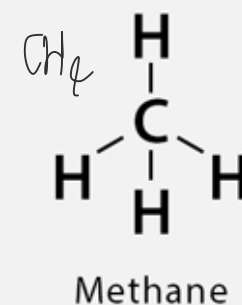
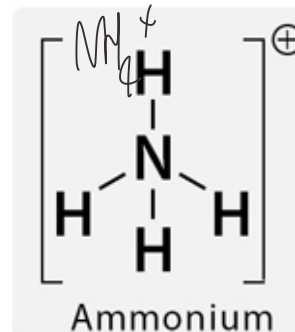
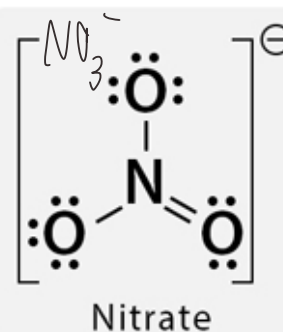
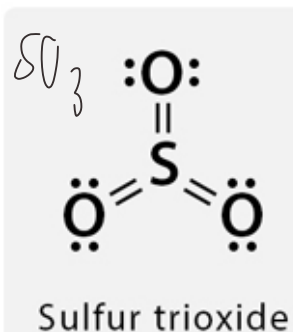
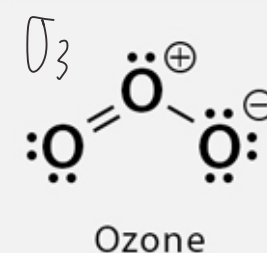
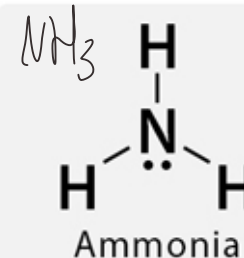
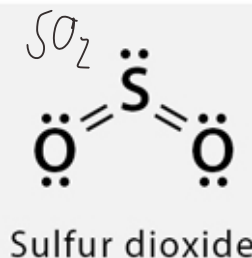
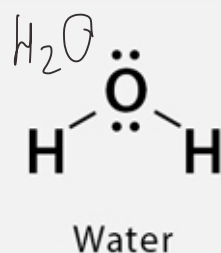
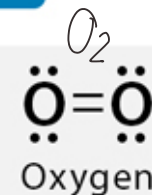
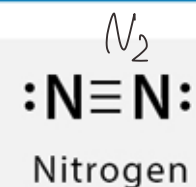
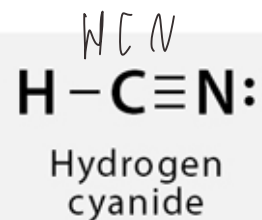
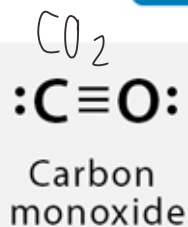
Compounds



ChemistryLearner.com

<https://www.chemistrylearner.com/lewis-structure.html>

Lewis Structure Examples



ChemistryLearner.com

<https://www.chemistrylearner.com/lewis-structure.html>



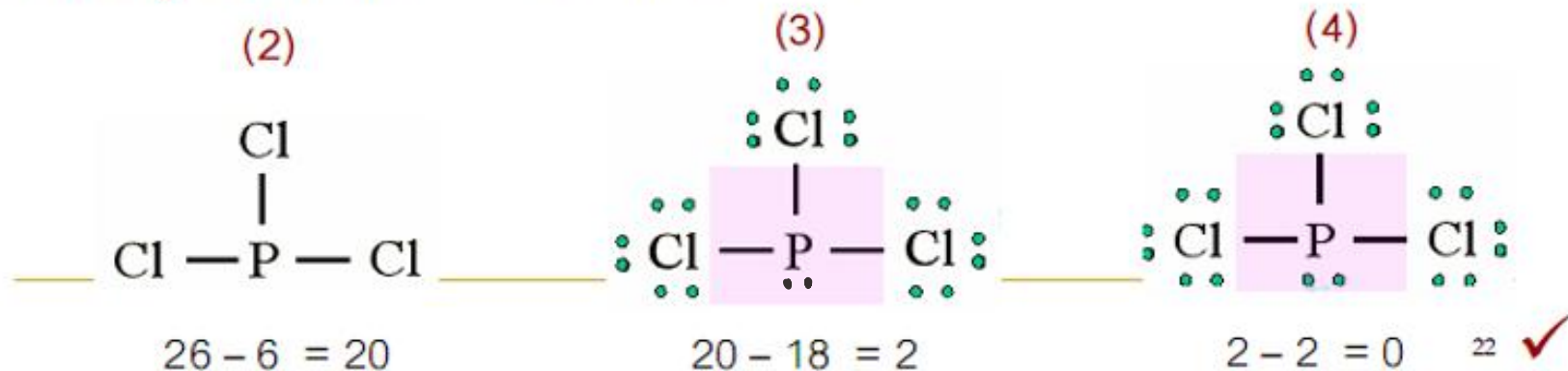
Lewis Structure

Steps for Writing Lewis Structures

- Step 1** Obtain the sum of the valence electrons from all of the atoms. Do not worry about keeping track of which electrons come from which atoms. It is the *total* number of valence electrons that is important.
- Step 2** Use one pair of electrons to form a bond between each pair of bound atoms. For convenience, a line (instead of a pair of dots) is often used to indicate each pair of bonding electrons.
- Step 3** Arrange the remaining electrons to satisfy the duet rule for hydrogen and the octet rule for each second-row element.

Writing Lewis Structures

Ex. PCl_3 (1) Total Ve- = $5 + (3 \times 7) = 26$





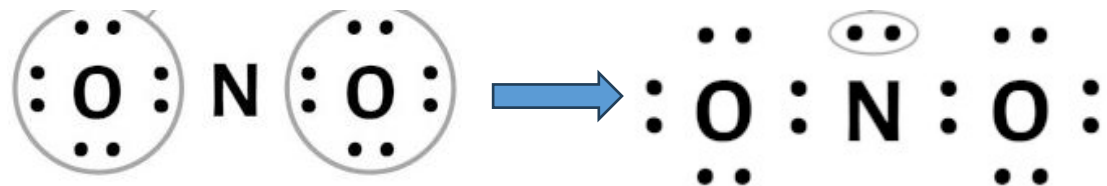
Ex. Write Lewis Structures of NO_2^-

(1) Total valence electrons (Ve^-) = $5 + (2 \times 6) + 1 = 18 \text{ Ve}^-$

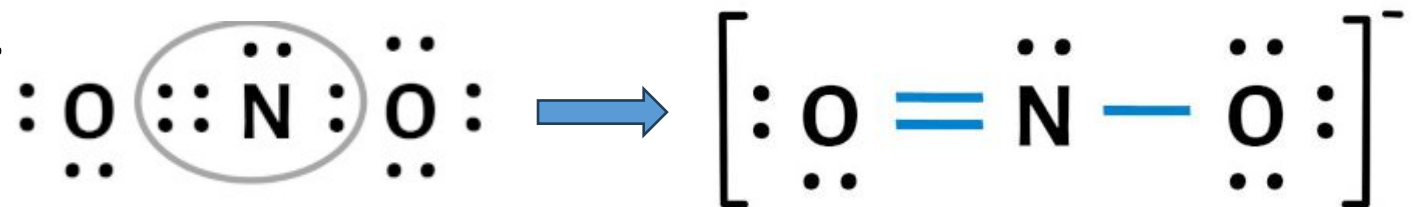
(2) Select the central atom/Connect each atoms by putting an electron pair between them



(3) Make the outer atoms stable/Place the remaining valence electrons pair on the central atom.



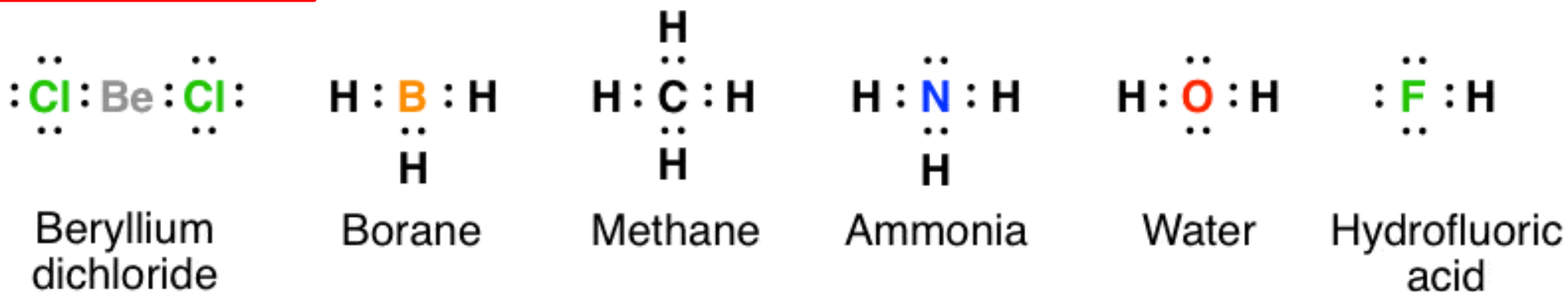
(4) Check the octet on the central atom. If it does not have octet, then shift the lone pair to form a double bond or triple bond.





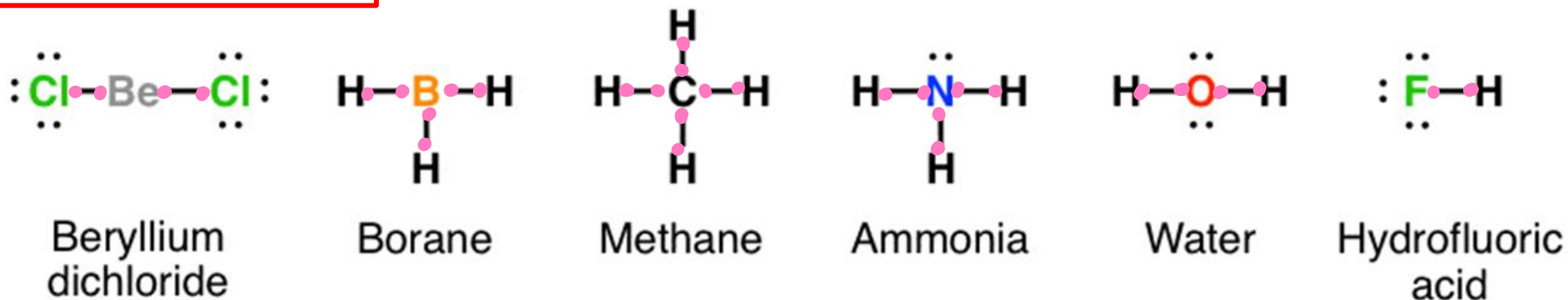
Lewis Structure

The Full Lewis (Lewis Dot Structure)



Vs.

The "Half-Lewis" (Lewis Dot + line Structures)



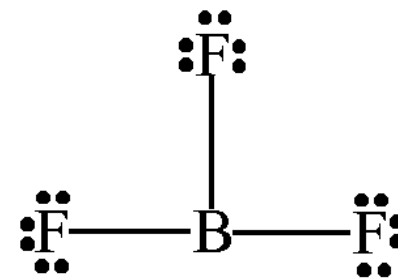


Some Exceptions to the Octet Rule

1. The Incomplete Octet

ไม่ครบ octet

- Fewer than eight :
- Group 2A (stable with 4 e⁻), BeH₂
- Group 3A (stable with 6 e⁻), BF₃

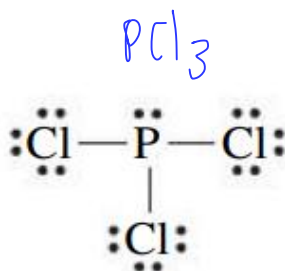
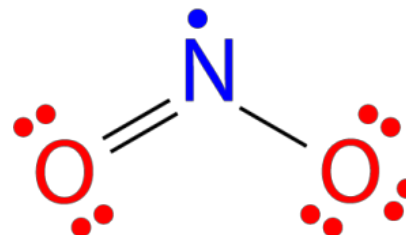


2. Odd-Electron Molecules: NO, NO₂

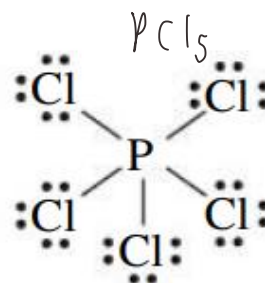
คี่ octet



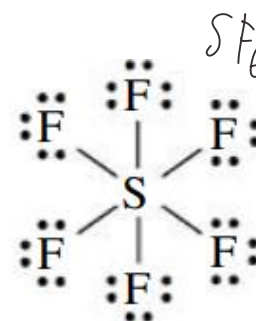
Nitric oxide



Octet



Expanded valence shell



Expanded valence shell

3. The Expanded Octet

ครบ

- Atoms of the second-period elements cannot have more than eight valence electrons around the central atom, but atoms of elements in and beyond the third period of the periodic table form some compounds in which more than eight electrons surround the central atom. SF₆, PCl₅, XeF₄



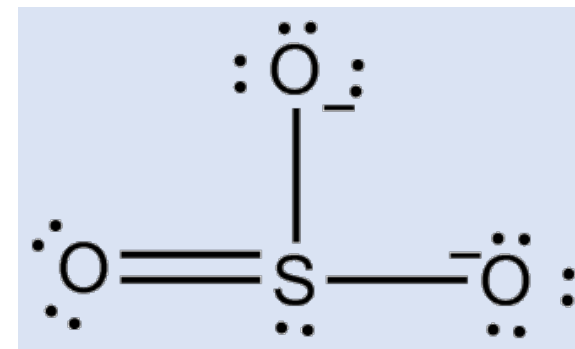
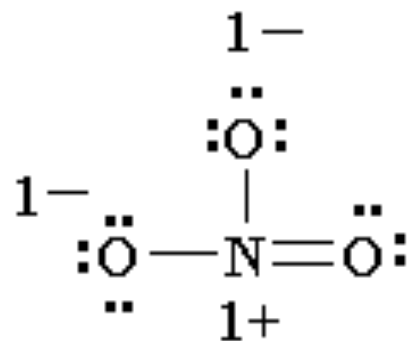
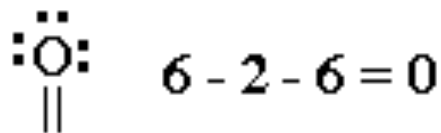
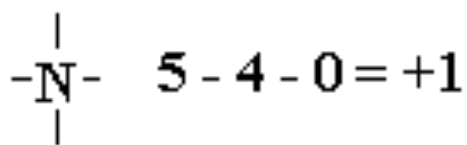
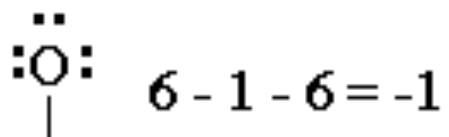
Formal Charge and Lewis Structure

Formal Charge is the electrical charge difference between the valence electrons in an isolated atom and the number of electrons assigned to the atom in a Lewis Structure.

ประจุต่าง

$$FC = [(\text{number valence } e^- \text{ in free atom}) - (\text{number lone-pair } e^-) - \frac{1}{2}(\text{number bond-pair } e^-)]$$

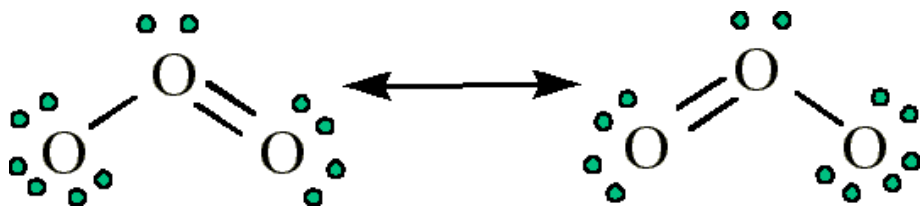
จล. อะตอมที่ว่าง - จล. e^- ที่ไม่ทำพันธะ - จล. คู่ที่พันธะ



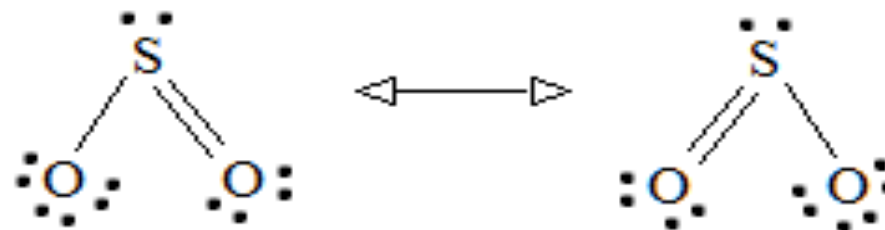
$$\begin{aligned} FC(S) &= 6 - 2 - \frac{1}{2}(8) \\ &= 6 - 2 - 4 = 0 \end{aligned}$$



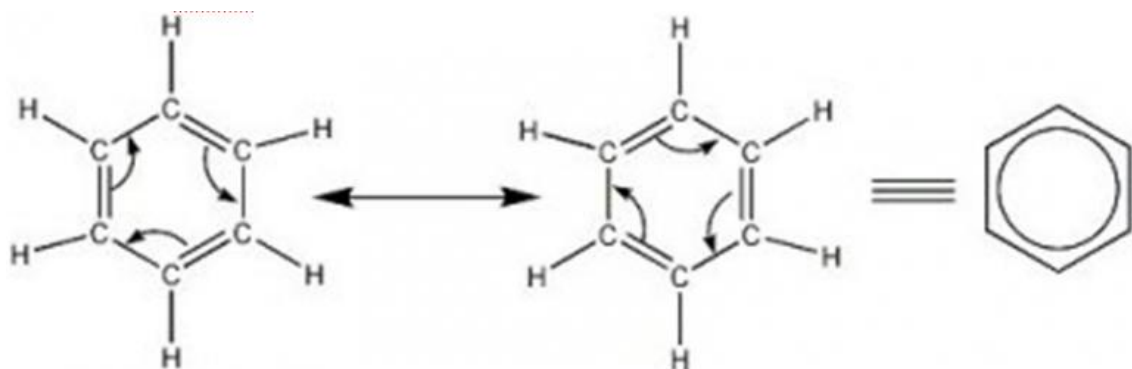
Resonance



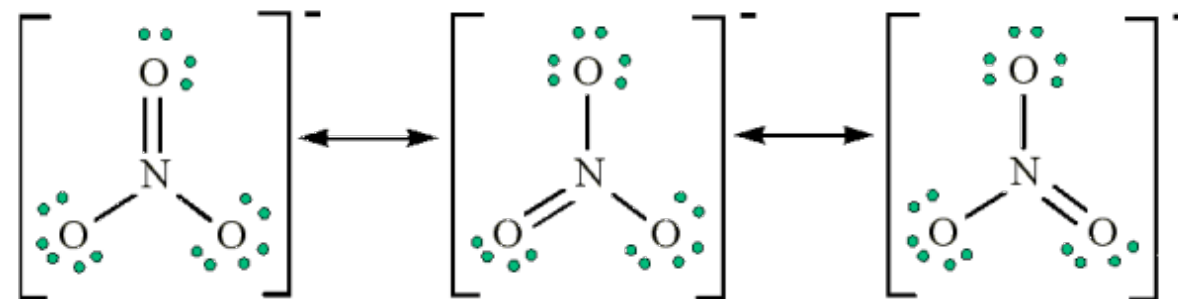
O_3



SO_3



Benzene



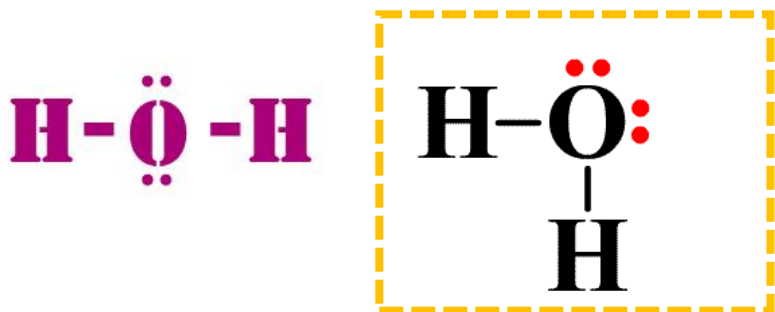
$[\text{NO}_3]^-$



Molecular Structure: The VSEPR Model

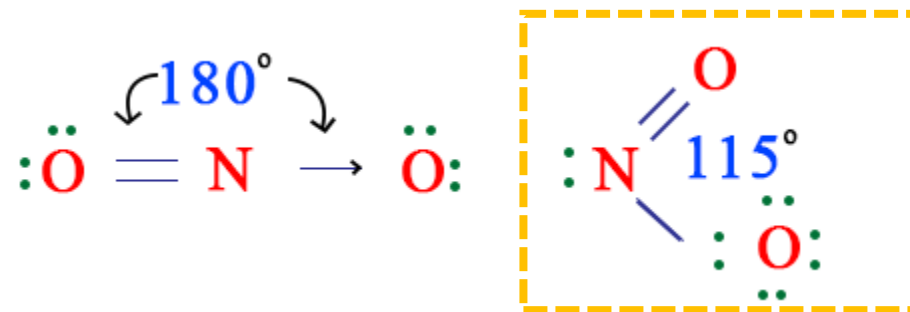
The **valence shell electron pair repulsion (VSEPR) model** is useful for predicting the molecular structures of molecules formed from nonmetals. The main idea of this model is that *the structure around a given atom is determined by minimizing repulsions between electron pairs*. This means that the bonding and nonbonding electron pairs (lone pairs) around a given atom are positioned *as far apart as possible*.

Lewis Structure



Molecular geometry
(Real structure)

Lewis Structure



Molecular geometry
(Real structure)



Number of electron pairs	Electron pair geometries: 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear AB_2				
3	 Trigonal planar AB_3E	 Bent or angular AB_2E_2			
4	 Tetrahedral AB_4	 Trigonal pyramidal AB_3E	 Bent or angular AB_2E_2		
5	 Trigonal bipyramidal AB_5	 Sawhorse or seesaw AB_4E	 T-shape AB_3E_2 <i>non polar</i>	 Linear AB_2E_3	
6	 Octahedral AB_6	 Square pyramidal AB_5E	 Square planar AB_4E_2 <i>non polar</i>	 T-shape AB_3E_3 <i>AX_2</i>	 Linear AB_2E_4

Electron Geometry	Linear	Trigonal planar	Tetrahedral	Trigonal bipyramidal	Octahedral
Spatial Arrangement					
Ball and Stick Model					
AX_m Notation	AX_2	AX_3	AX_4	AX_5	AX_6
Line-dash-wedge Notation					

Four electron groups	Electron-group structure			
	Molecular shape			
		Tetrahedral	Pyramidal	Bent
Three electron groups	Electron-group structure			
	Molecular shape			
Two electron groups	Electron-group structure			
	Molecular shape			

Key:
 Lone pair
 Bonding electron group



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Formula	Example	Geometry	Polyhedron	Ball & Stick Model
AB_2	$H-Be-H$	Linear	Line	
AB_3		Trigonal Planar	Triangle	
AB_2e	$H-\ddot{C}-H$	Bent	Triangle	

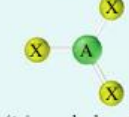
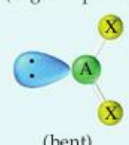
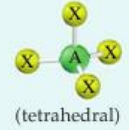
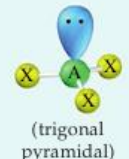
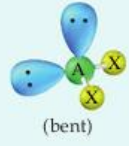
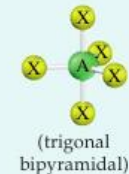
AB_5		Trigonal Bipyramidal	Trigonal Bipyramid	
AB_4e		Seesaw	Trigonal Bipyramid	
AB_3e_2		T-shaped	Trigonal Bipyramid	
AB_2e_3	$H-Xe-H$	Linear	Trigonal Bipyramid	

Formula	Example	Geometry	Polyhedron	Ball & Stick Model
AB_4	$H-Si-H$	Tetrahedral	Tetrahedron	
AB_3e	$H-\ddot{N}-H$	Pyramidal	Tetrahedron	
AB_2e_2		Bent	Tetrahedron	

Formula	Example	Geometry	Polyhedron	Ball & Stick Model
AB_6		Octahedral	Octahedron	
AB_5e		Square Pyramidal	Octahedron	
AB_4e_2		Square Planar	Octahedron	

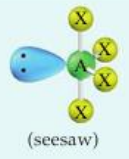
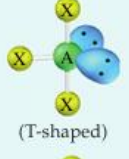
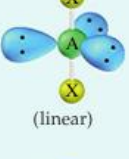
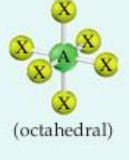
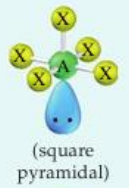
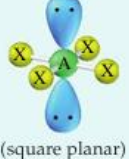


TABLE 10.1 Molecular Geometry as a Function of Electron-Group Geometry

Number of Electron Groups	Electron-Group Geometry	Number of Lone Pairs	VSEPR Notation	Molecular Geometry	Ideal Bond Angles	Example
2	linear	0	AX ₂	 (linear)	180°	BeCl ₂ 
3	trigonal planar	0	AX ₃	 (trigonal planar)	120°	BF ₃ 
	trigonal planar	1	AX ₂ E	 (bent)	120°	SO ₂ ^a 
4	tetrahedral	0	AX ₄	 (tetrahedral)	109.5°	CH ₄ 
	tetrahedral	1	AX ₃ E	 (trigonal pyramidal)	109.5°	NH ₃
	tetrahedral	2	AX ₂ E ₂	 (bent)	109.5°	OH ₂
5	trigonal bipyramidal	0	AX ₅	 (trigonal bipyramidal)	90°, 120°	PCl ₅ 

(continued)

TABLE 10.1 Molecular Geometry as a Function of Electron-Group Geometry (Continued)

Number of Electron Groups	Electron-Group Geometry	Number of Lone Pairs	VSEPR Notation	Molecular Geometry	Ideal Bond Angles	Example
	trigonal bipyramidal	1	AX ₄ E ^b	 (seesaw)	90°, 120°	SF ₄
	trigonal bipyramidal	2	AX ₃ E ₂	 (T-shaped)	90°	ClF ₃
	trigonal bipyramidal	3	AX ₂ E ₃	 (linear)	180°	XeF ₂
6	octahedral	0	AX ₆	 (octahedral)	90°	SF ₆ 
	octahedral	1	AX ₅ E	 (square pyramidal)	90°	BrF ₅
	octahedral	2	AX ₄ E ₂	 (square planar)	90°	XeF ₄

^aFor a discussion of the structure of SO₂, see page 428.

^bFor a discussion of the placement of the lone-pair electrons in this structure, see page 427.



Chemical bonds/Chemical bonding

Ref: Ralph H. Petrucci, F. Geoffrey Herring, Jeffry D. Madura, Carey Bissonnette. General Chemistry: Principles and Modern Applications. Prentice Hall 2010, English 10th ed.

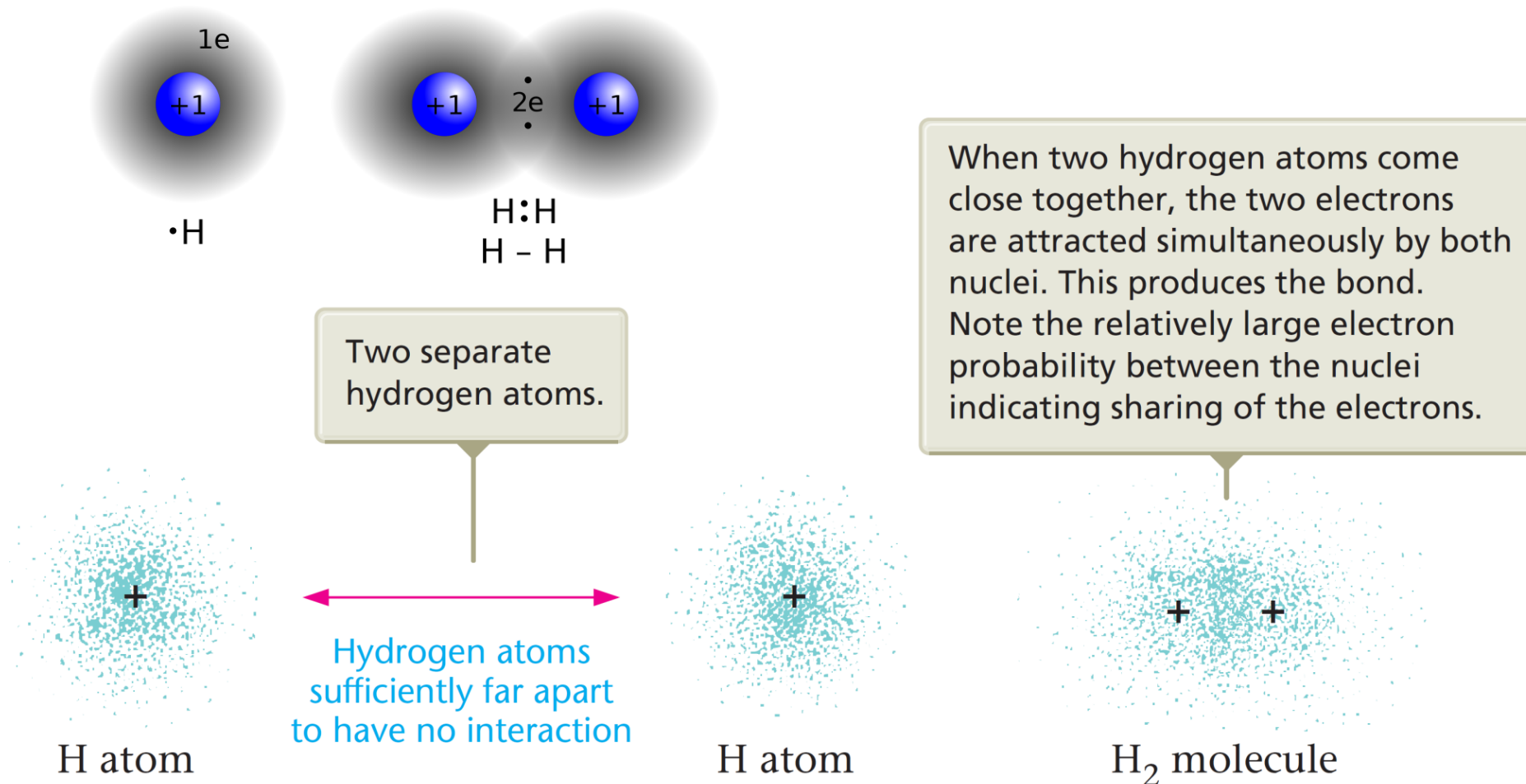
Theodore E. Brown, H. Eugene H LeMay, Bruce E. Bursten, Catherine Murphy, Patrick Woodward. Chemistry: The Central Science. Prentice Hall 2011, English 12th ed.

Raymond Chang and Jason Overby. General Chemistry: The Essential Concepts. McGraw-Hill Science/Engineering/Math 2010, 6th ed.



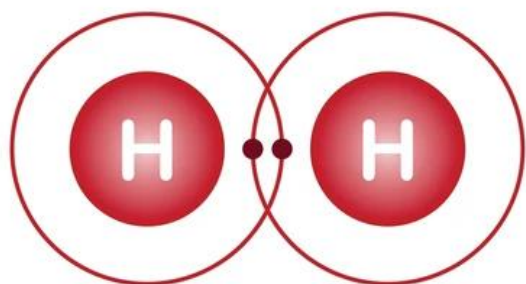
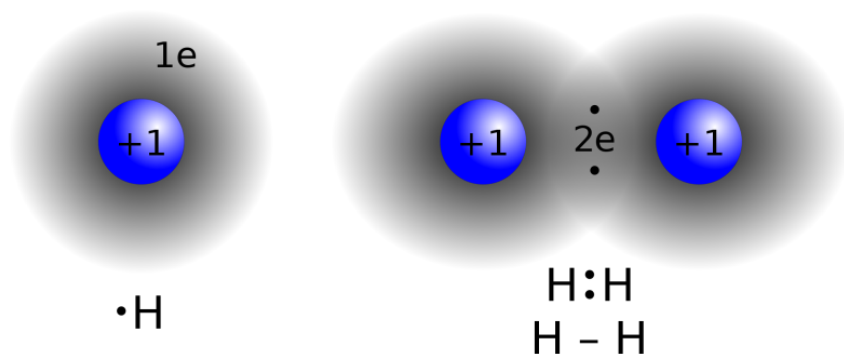
Covalent Bond

Electrons are shared by nuclei is called **covalent bonding**

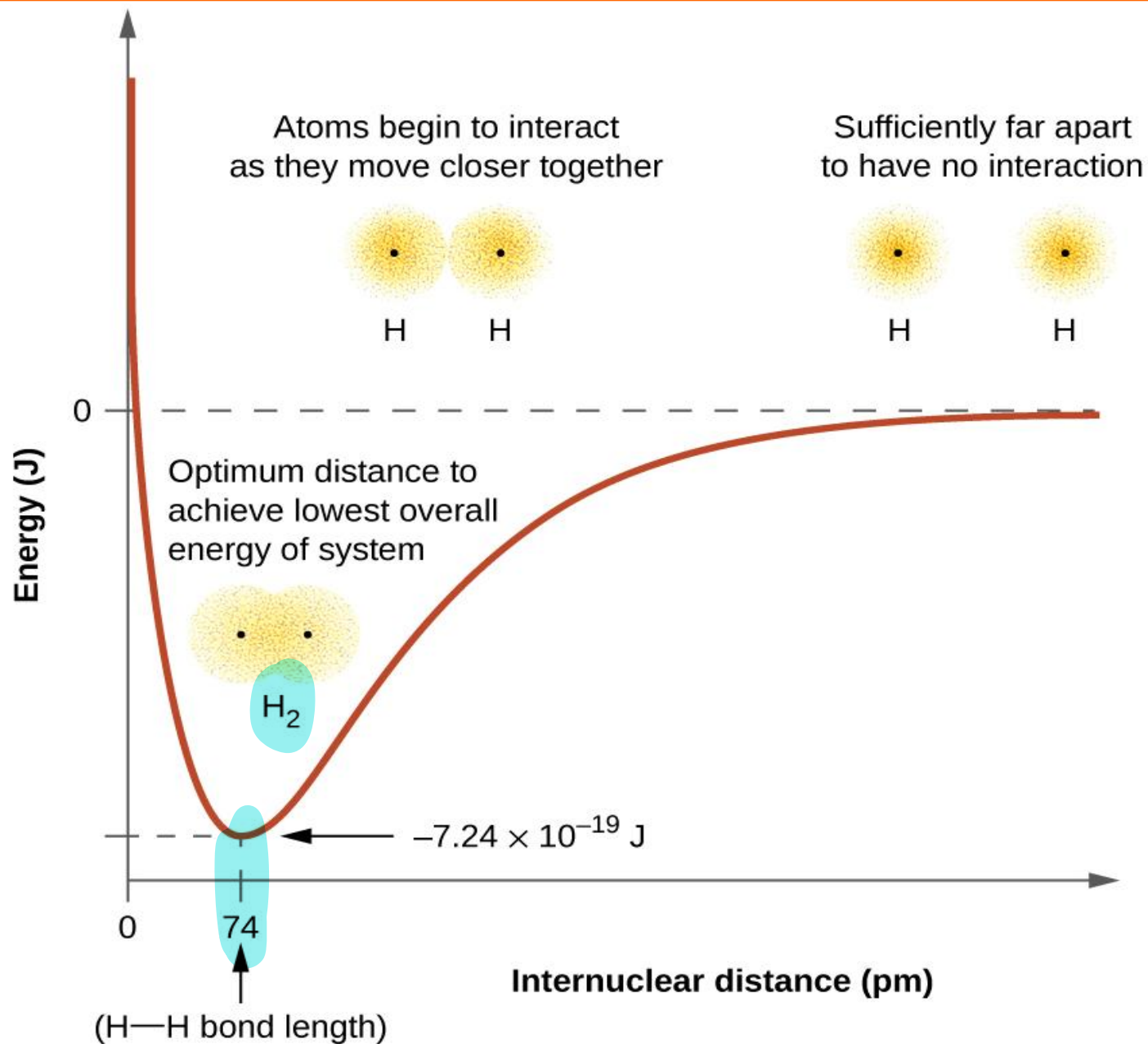




Covalent Bonding



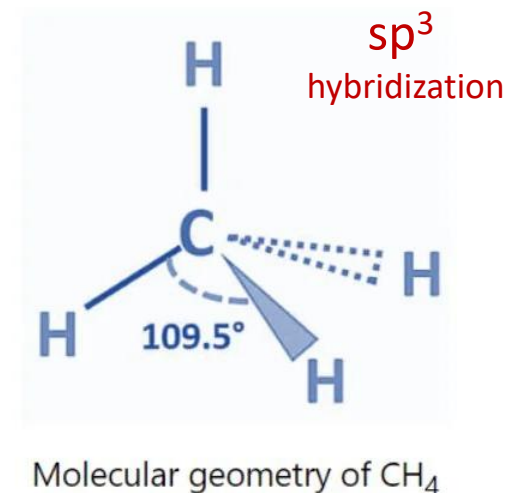
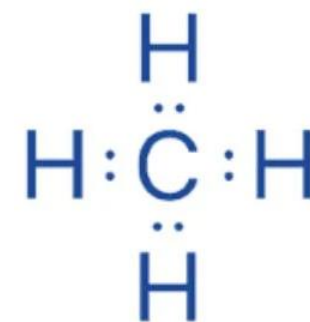
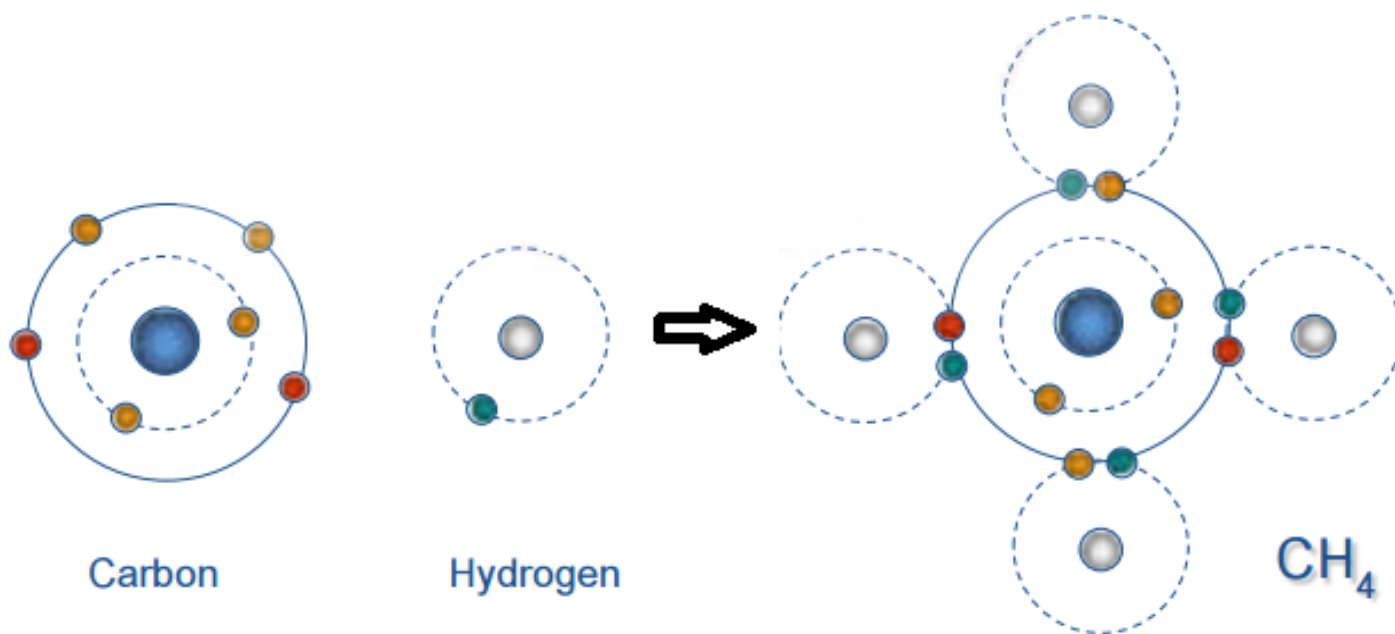
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Chemical Bonds - Covalent Bonding

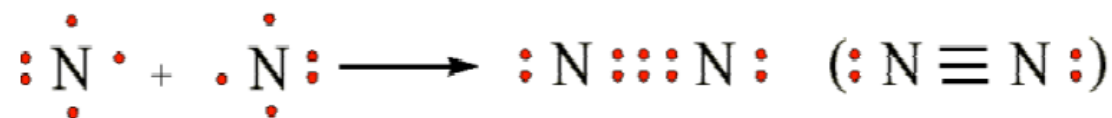
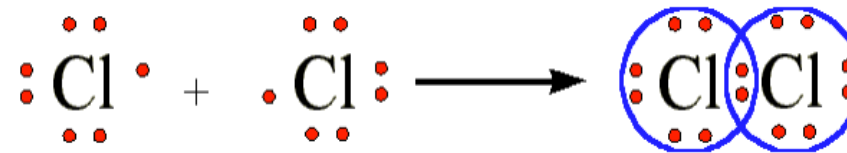
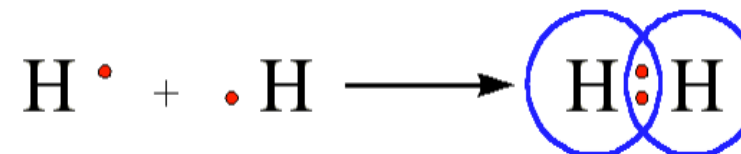
Lewis Structures of CH₄



bonding electron

Vs.

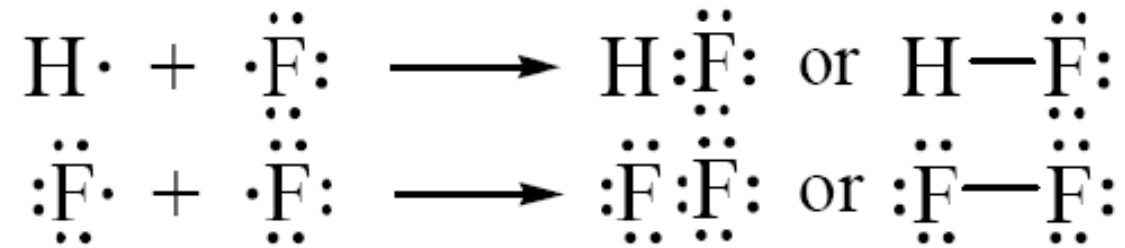
non-bonding electron



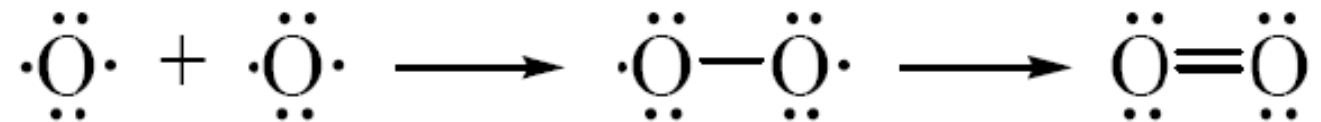


Chemical Bonds – Types of Covalent bonds

- *Single Covalent Bonds*



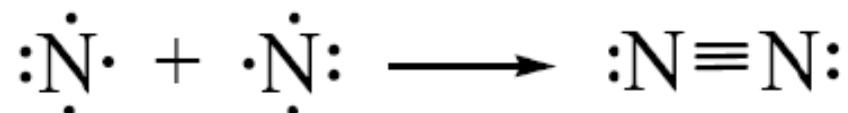
- *Double Covalent Bonds*



bonding electron

- *Triple Covalent Bonds*

non-bonding electron



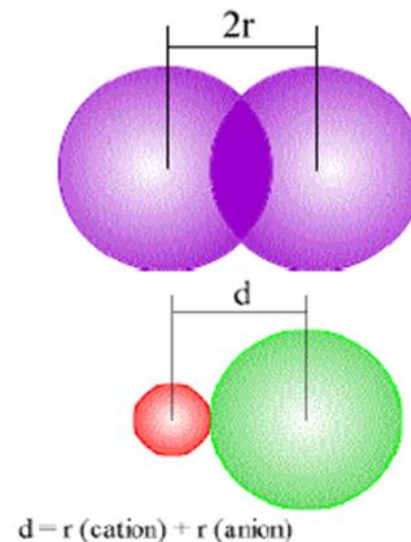
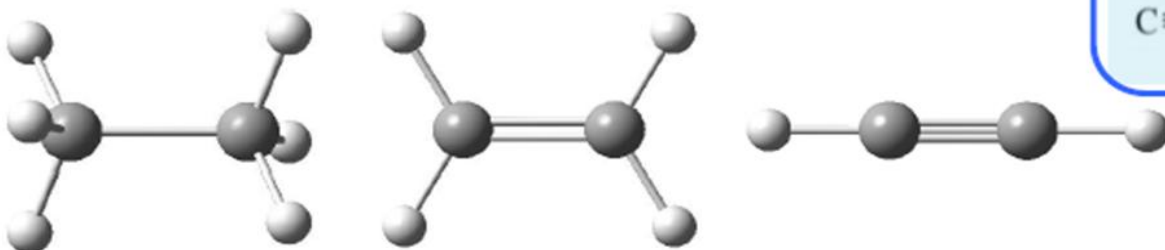


Bond Lengths-ความยาวพันธะ

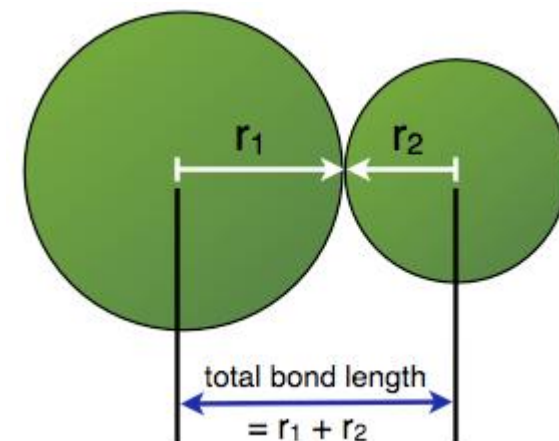
ความยาวพันธะ คือระยะห่างระหว่างอะตอมคู่ที่สร้างพันธะ โดยเป็นตำแหน่งที่อะตอมทั้งสองดึงดูดกันได้ดีที่สุด มีพลังงานต่ำสุดหรือมีเสถียรภาพที่สุด

○ ความยาวของพันธะโควาเลนต์สัมพันธ์กับพลังงานพันธะ

- ความยาวพันธะเดี่ยว > พันธะคู่ > พันธะสาม
- พลังงานพันธะเดี่ยว < พันธะคู่ < พันธะสาม



	Bond Length	Bond Energy
C—C	154	347
C=C	134	611
C≡C	120	837





Bond Lengths

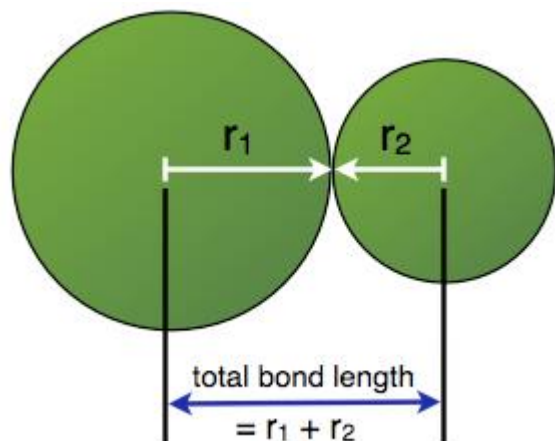
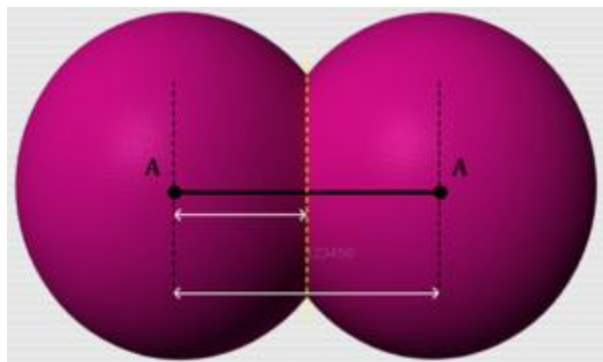


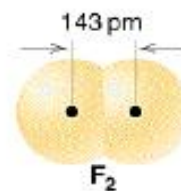
TABLE 11.2 Some Average Bond Lengths^a

Bond	Bond Length, pm	Bond	Bond Length, pm	Bond	Bond Length, pm
H—H	74.14	C—C	154	N—N	145
H—C	110	C=C	134	N=N	123
H—N	100	C≡C	120	N≡N	109.8
H—O	97	C—N	147	N—O	136
H—S	132	C=N	128	N=O	120
H—F	91.7	C≡N	116	O—O	145
H—Cl	127.4	C—O	143	O=O	121
H—Br	141.4	C=O	120	F—F	143
H—I	160.9	C—Cl	178	Cl—Cl	199
				Br—Br	228
				I—I	266

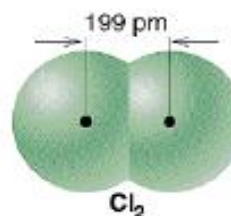


Internuclear distance
(bond length)

Covalent
radius



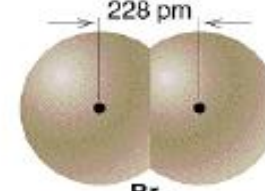
F_2



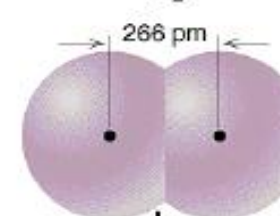
Cl_2

Internuclear distance
(bond length)

Covalent
radius



Br_2



I_2



Bond Energy or Bond enthalpy

is the measure of bond strength in a chemical bond, **IUPAC** defines **bond energy as the average value** of the gas-phase bond-dissociation energy

TABLE 11.3 Some Average Bond Energies^a

Bond	Bond Energy, kJ/mol	Bond	Bond Energy, kJ/mol	Bond	Bond Energy, kJ/mol
H—H	436	C—C	347	N—N	163
H—C	414	C=C	611	N=N	418
H—N	389	C≡C	837	N≡N	946
H—O	464	C—N	305	N—O	222
H—S	368	C=N	615	N=O	590
H—F	565	C≡N	891	O—O	142
H—Cl	431	C—O	360	O=O	498
H—Br	364	C=O	736 ^b	F—F	159
H—I	297	C—Cl	339	Cl—Cl	243
				Br—Br	193
				I—I	151

Average Bond Energy

■ Average Bond Energies

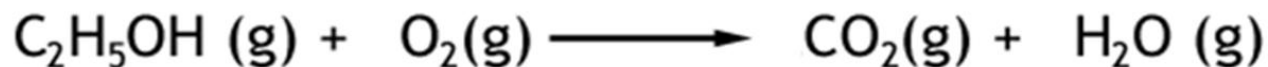
Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
H—H	436	N—N	163	Br—F	237
H—C	414	N=N	418	Br—Cl	218
H—N	389	N≡N	946	Br—Br	193
H—O	464	N—O	222	I—Cl	208
H—S	368	N=O	590	I—Br	175
H—F	565	N—F	272	I—I	151
H—Cl	431	N—Cl	200	Si—H	323
H—Br	364	N—Br	243	Si—Si	226
H—I	297	N—I	159	Si—C	301
C—C	347	O—O	142	S—O	265
C=C	611	O=O	498	Si=O	368
C≡C	837	O—F	190	S=O	523
C—N	305	O—Cl	203	Si—Cl	464
C=N	615	O—I	234	S=S	418
C≡N	891	F—F	159	S—F	327
C—O	360	Cl—F	253	S—Cl	253
C=O	736*	Cl—Cl	243	S—Br	218
C≡O	1072			S—S	266
C—Cl	339				

*799 in CO₂

<https://www.chegg.com/homework-help/questions-and-answers/bond-table-103-average-bond-energies-bond-energy-bond-energy-bond-kj-mol-bond-kj-mol-h-436-q63962316>



Ex. Heat of Reaction (enthalpy)



B(X-Y) / kJ mol⁻¹

H-C = 413	O-H = 463
C-C = 348	O=O = 498
C-O = 743	C=O = 743

$$\begin{aligned} \Delta H &= [(5 \times 413) + 348 + 360 + 463 + \underline{3} \times 498] \\ &\quad + [\underline{4} \times -743) + \underline{6} \times -463)] \\ &= -1020 \text{ kJ mol}^{-1} \end{aligned}$$

$$\Delta H_{\text{rxn}} = \sum(\text{bonds broken}) - \sum(\text{bonds formed})$$

$\Delta H = \sum(\text{bonds broken}) - \sum(\text{bonds formed})$

⊖ exo
⊕ endo

$$\Delta H_{\text{rxn}} = \sum(\text{BE}_{\text{reactants}}) - \sum(\text{BE}_{\text{products}})$$

(BE = Bond Energies)



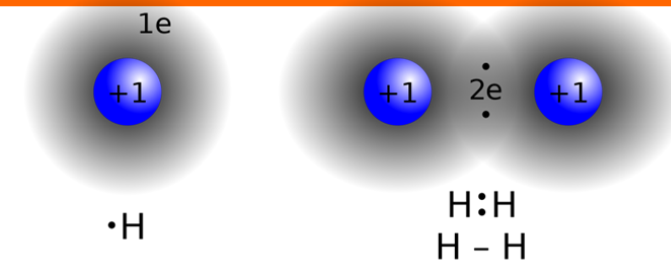
Exothermic → $-\Delta H^0$ ($\Delta H^0 < 0$)

Endothermic → $+\Delta H^0$ ($\Delta H^0 > 0$)



Covalent Bond

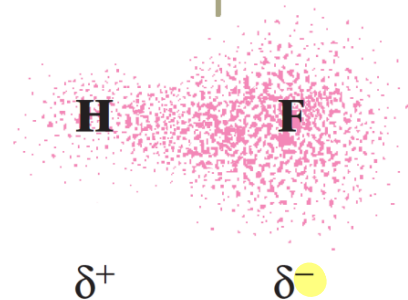
Unequal sharing of electrons results **polar covalent bond**.



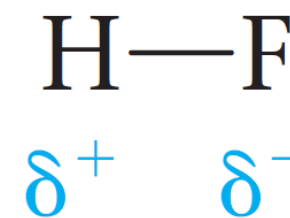
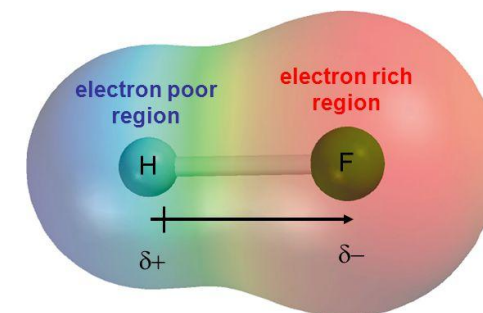
What the probability map would like if the two electrons in the H—F bond were shared equally.

**a**

The actual situation, where the shared pair spends more time **close to the fluorine atom than to the hydrogen atom**. This gives fluorine a slight excess of negative charge and the hydrogen a slight deficit of negative charge (a slight positive charge).

**b**

Polar Molecules



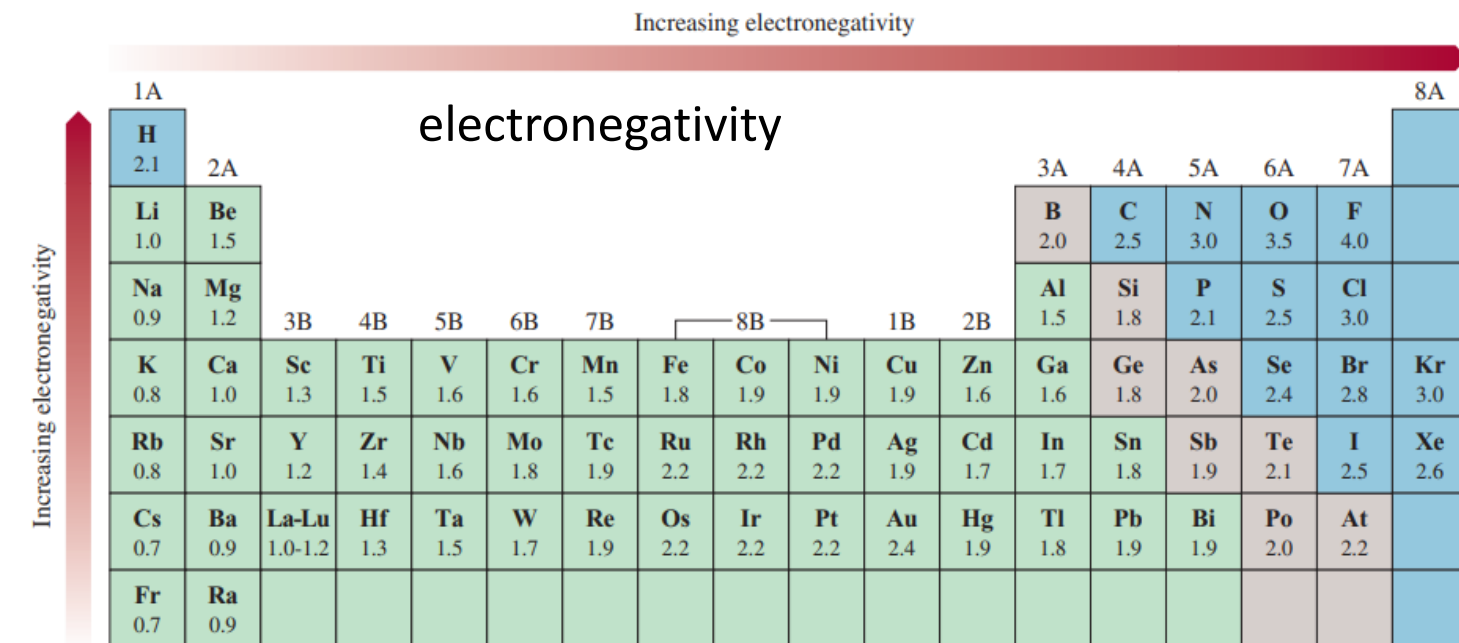
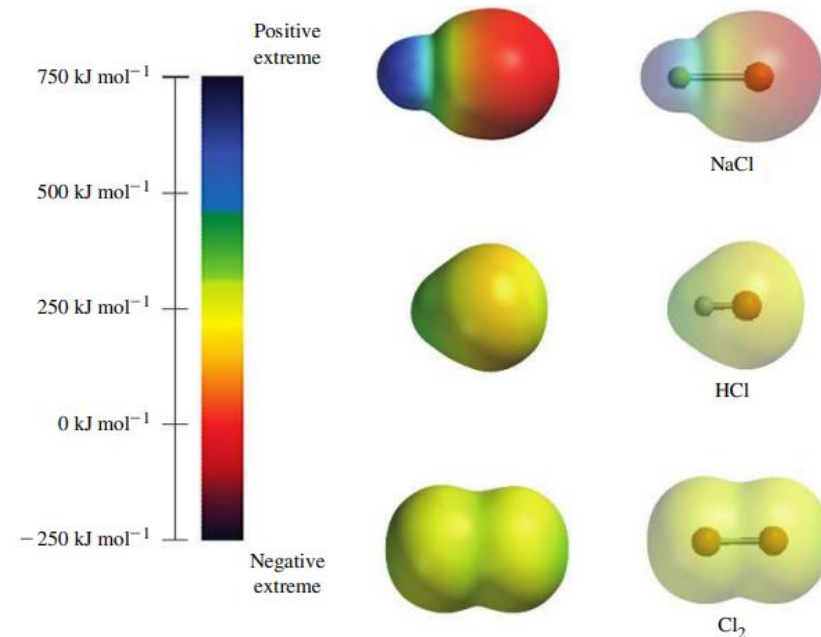
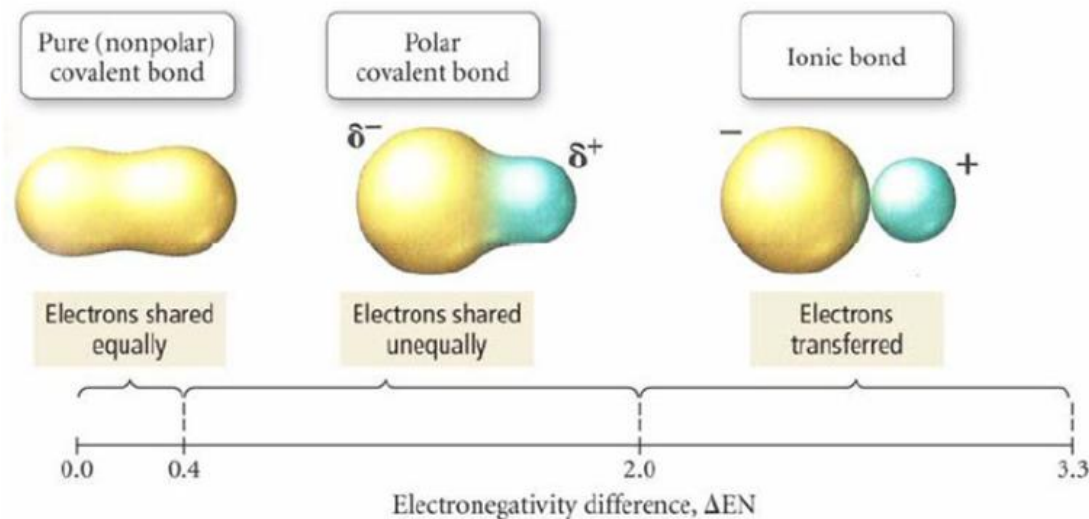
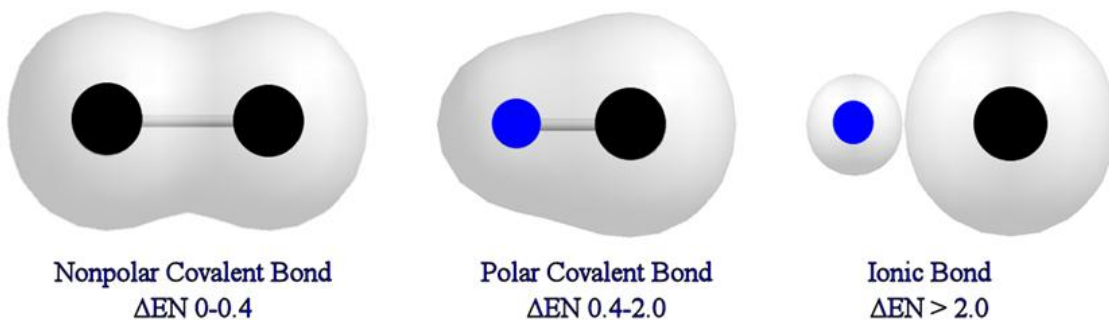


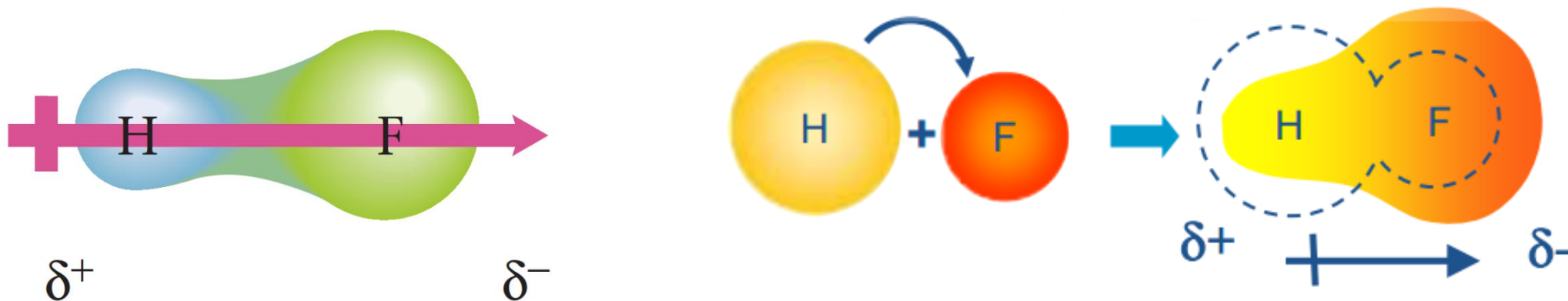
Figure 9.5
The electronegativities of common elements.





Bond Polarity and Dipole Moments

A molecule such as HF that has a center of positive charge and a center of negative charge is said to have a **dipole moment**. The dipolar character of a molecule is often represented by an arrow. This arrow points toward the negative charge center, and its tail indicates the positive center of charge.





Bond Polarity and Dipole Moments

1. ^{ไม่ขั้ว} Non-polar covalent bond



A nonpolar covalent bond

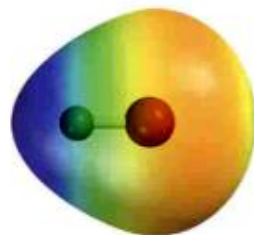


2. ^{ขั้ว} Polar covalent bond

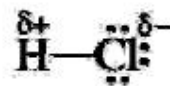


A polar covalent bond.

The bonding electrons are attracted more strongly by Cl than by H.



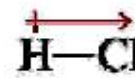
(a)



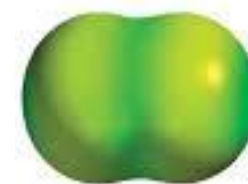
(c)



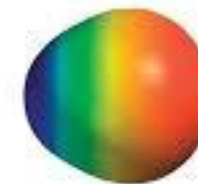
(b)



(d)



F₂



HF



LiF



F-F



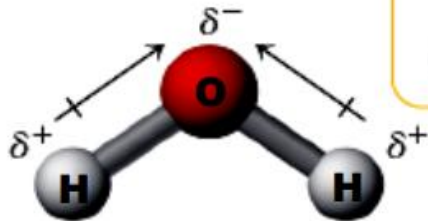
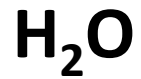
$\delta^+ \delta^-$
H-F



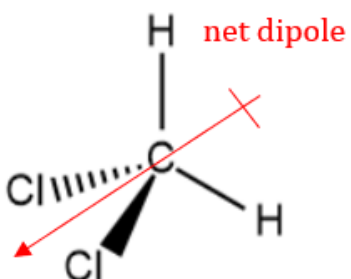
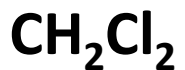
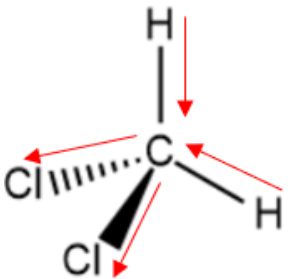
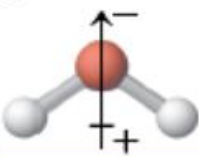
$^+ ^-$
Li F



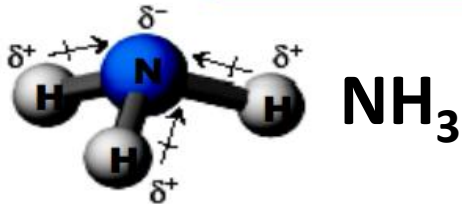
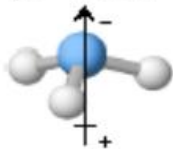
Bond Polarity and Dipole Moments



Net dipole
 $\mu = 1.85\text{D}$

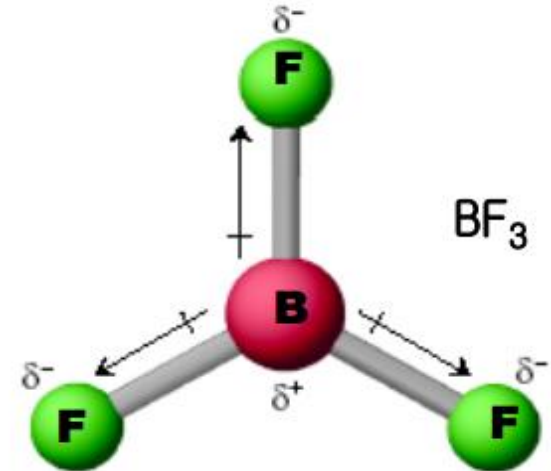
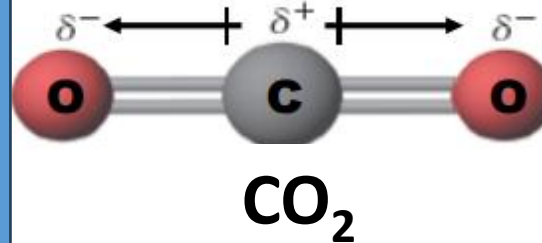


Net dipole
 $\mu = 1.47\text{D}$



Polar Covalent

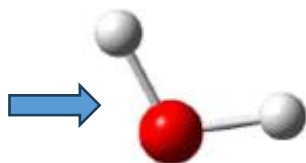
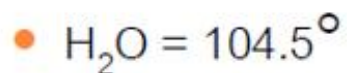
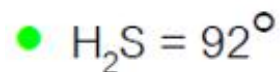
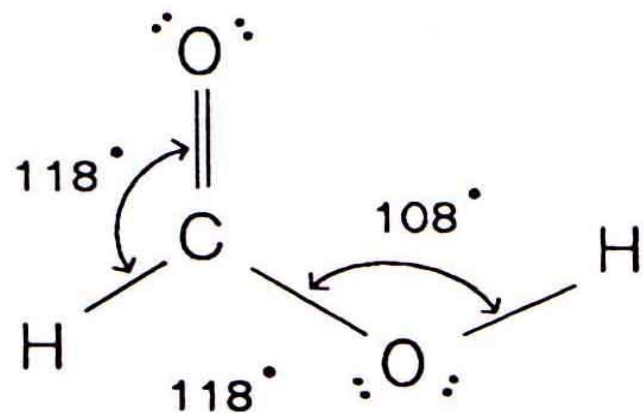
No net dipole
moment



Non-Polar Covalent



Bond Angle



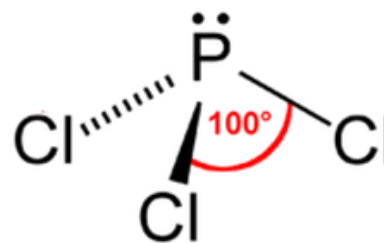
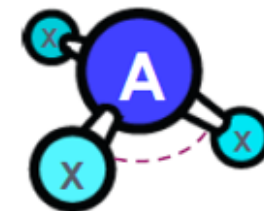
Bond Angle



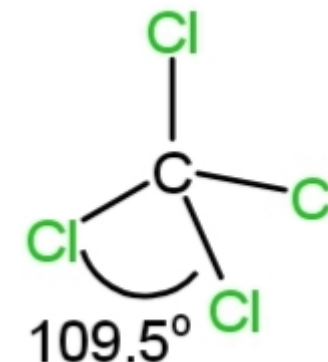
Electronegativity and atomic size effects

AX_3	PI_3	PBr_3	PCl_3	PF_3
EN(X)	2.66	2.96	3.16	3.98
angle	102.0	101.5	100.3	97.8

AX_3	AsH_3	PH_3	NH_3
EN(A)	2.17	2.19	3.04
angle	91.6	93.8	107.0



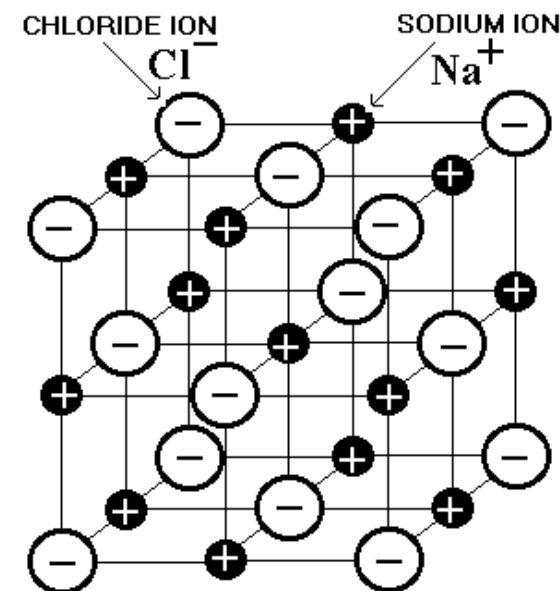
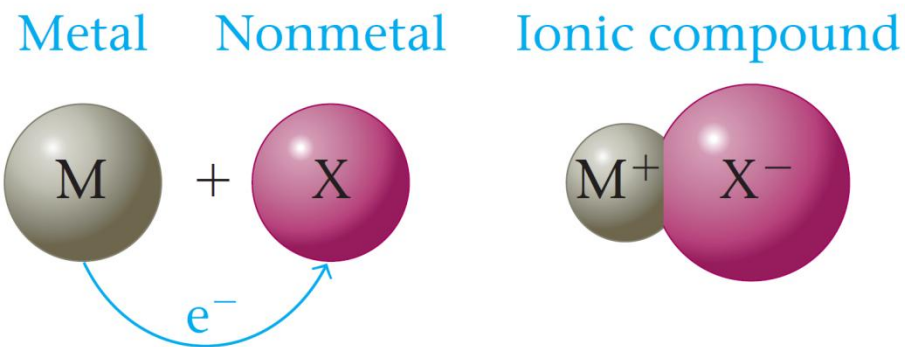
Vs.





Ionic Bond

Ionic substances are formed when an atom that **loses electrons** relatively easily reacts with an atom that has a high affinity for electrons. In other words, an **ionic compound** results when a metal reacts with a nonmetal.

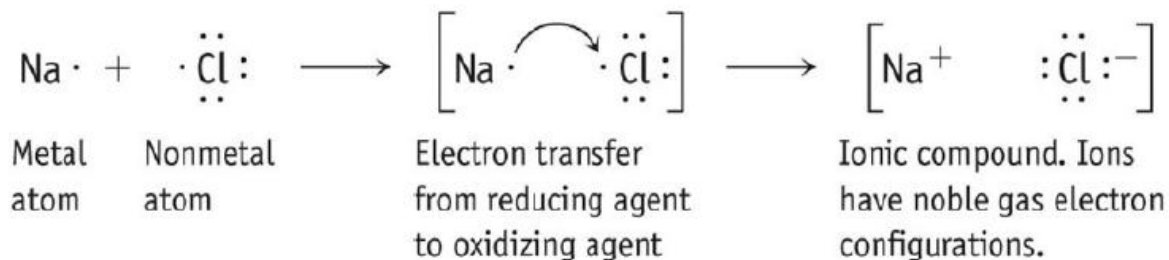


In almost **all stable chemical compounds** of the representative elements, all of the atoms have **achieved a noble gas electron configuration**



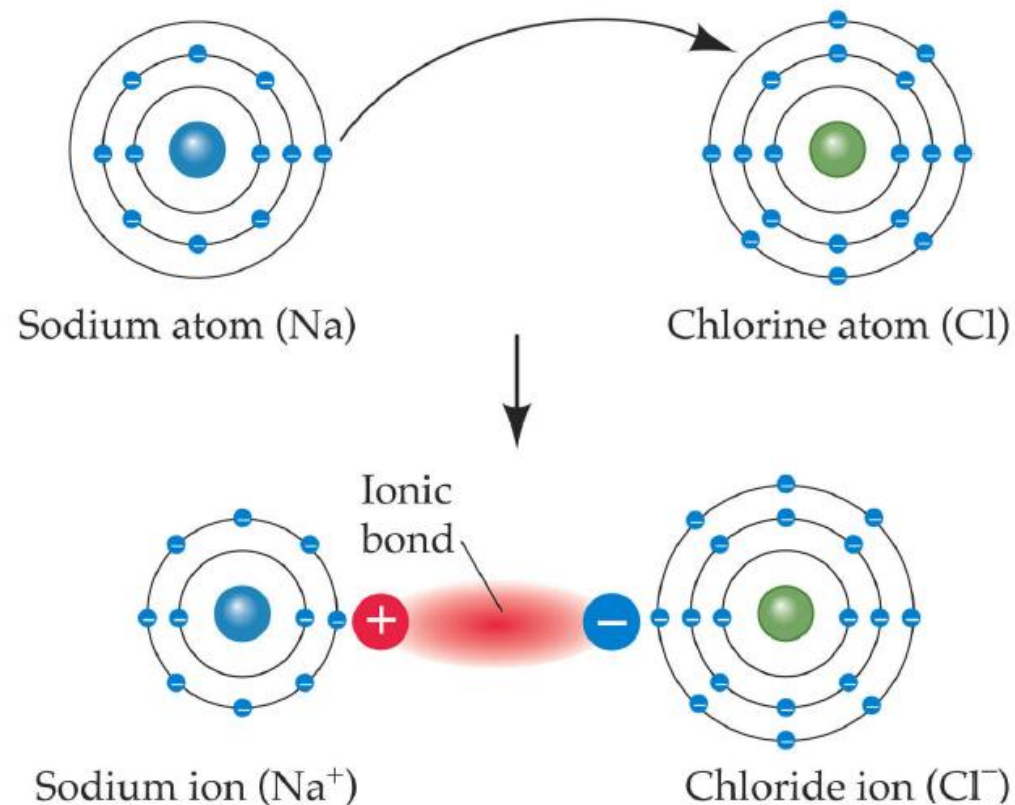
Ionic Bond

Forming an ionic compound



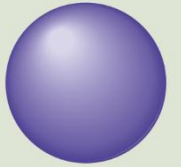
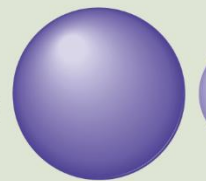
The Na^+ and Cl^- are held together by the electrostatic attraction of the two ions.

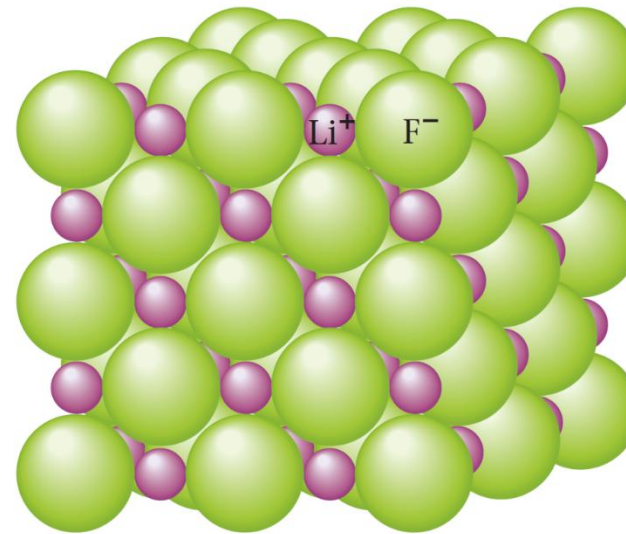
This electrostatic attraction is the **ionic bond**





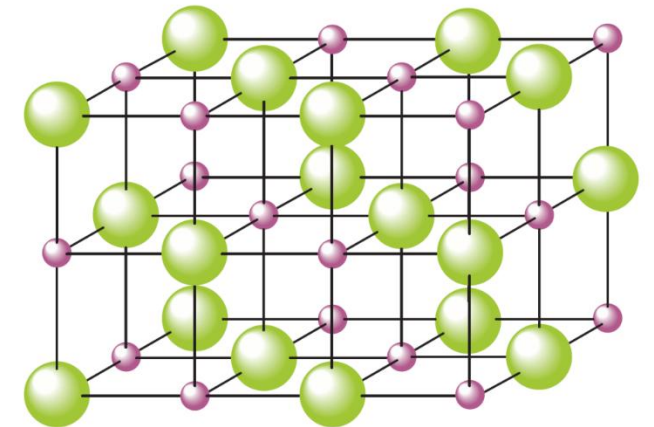
Ionic Bonding and Structures of Ionic Compounds

Atom		Cation		Atom		Anion	
Li 152			Li ⁺ 60	F 72			F ⁻ 136
Na 186			Na ⁺ 95	Cl 99			Cl ⁻ 181
K 227			K ⁺ 133	Br 114			Br ⁻ 195
Rb 248			Rb ⁺ 148	I 133			I ⁻ 216
Cs 265			Cs ⁺ 169				



a

This structure represents the ions as packed spheres.



b

This structure shows the positions (centers) of the ions. The spherical ions are packed in the way that maximizes the ionic attractions.

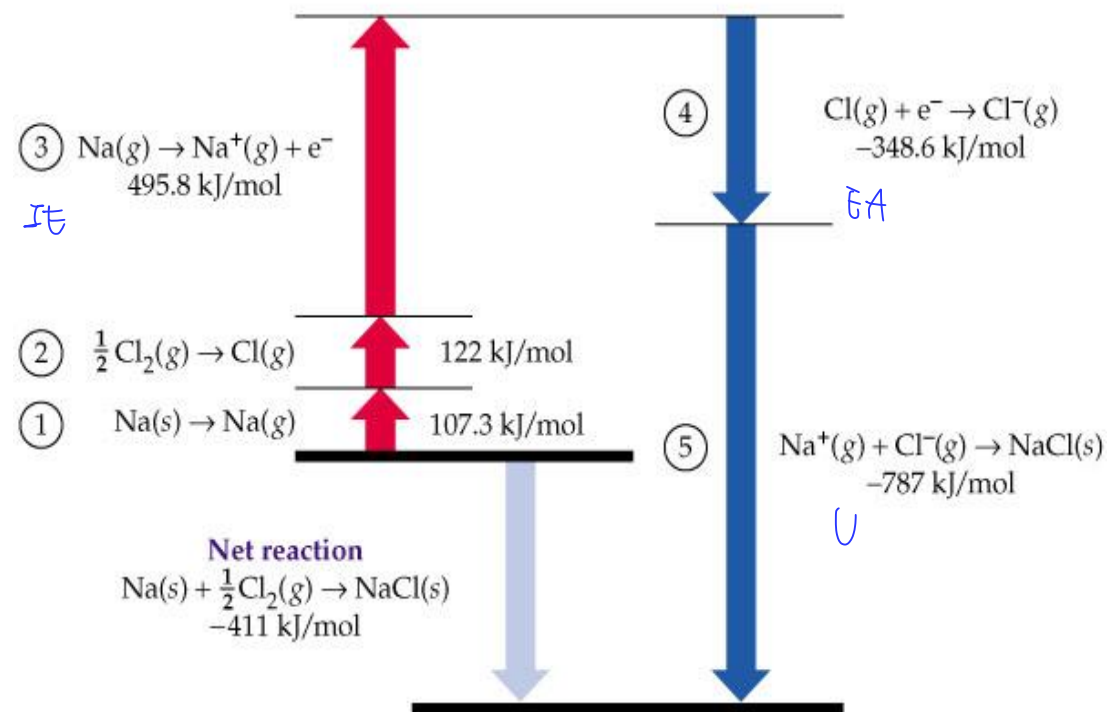
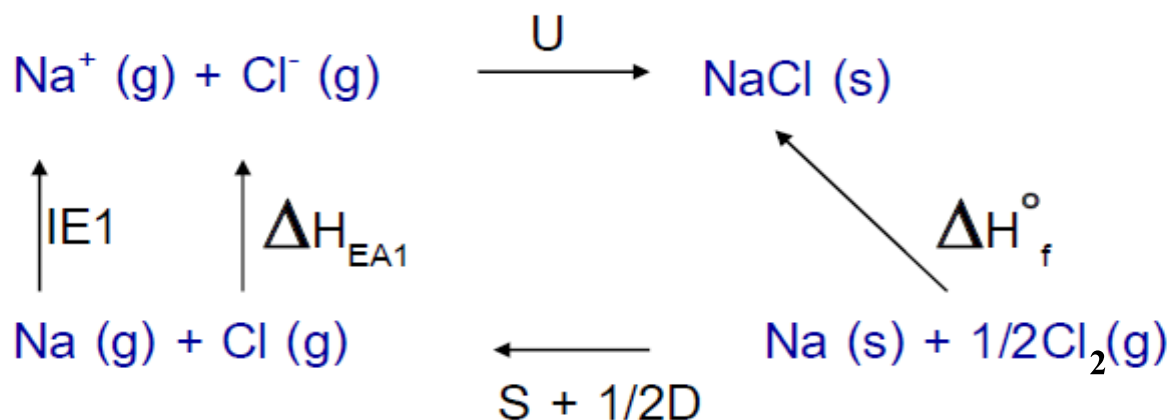


Lattice Energy: Born-Haber cycle

$$\Delta H_f^\circ = \Delta H_{\text{sub}} + \text{IE} + \Delta H_{\text{diss}} + \text{EA} + U$$

$$\Delta H_f^\circ = 108 + 496 + 122 - 349 - 788 = -411 \text{ kJ/mole}$$

• Born-Harber cycle



Sodium



Chlorine

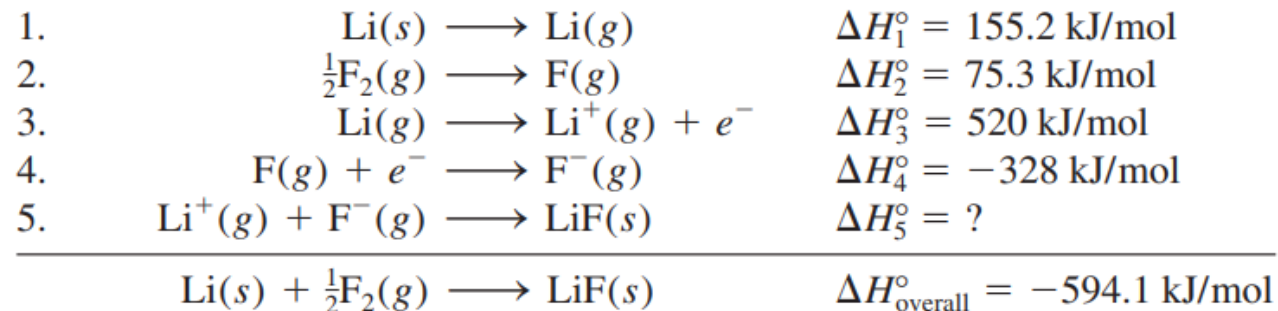
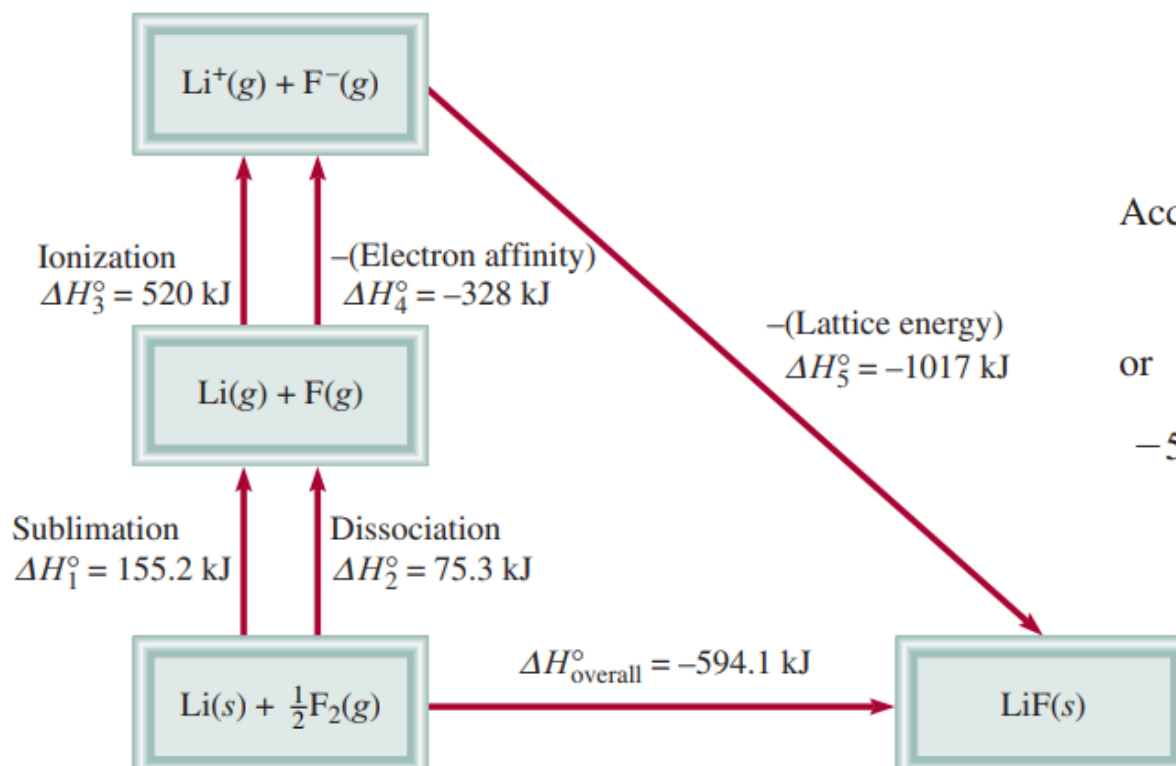


Sodium chloride

$$U \text{ (Lattice energy)} = \Delta H_f^\circ - S - 1/2 D - \text{IE}_1 - \Delta H_{\text{EA}}$$



The Born-Haber Cycle for Determining Lattice Energies



According to Hess's law, we can write

$$\Delta H_{\text{overall}}^\circ = \Delta H_1^\circ + \Delta H_2^\circ + \Delta H_3^\circ + \Delta H_4^\circ + \Delta H_5^\circ$$

or

$$-594.1 \text{ kJ/mol} = 155.2 \text{ kJ/mol} + 75.3 \text{ kJ/mol} + 520 \text{ kJ/mol} - 328 \text{ kJ/mol} + \Delta H_5^\circ$$

Lattice Energy of LiF

$$\Delta H_5^\circ = -1017 \text{ kJ/mol}$$

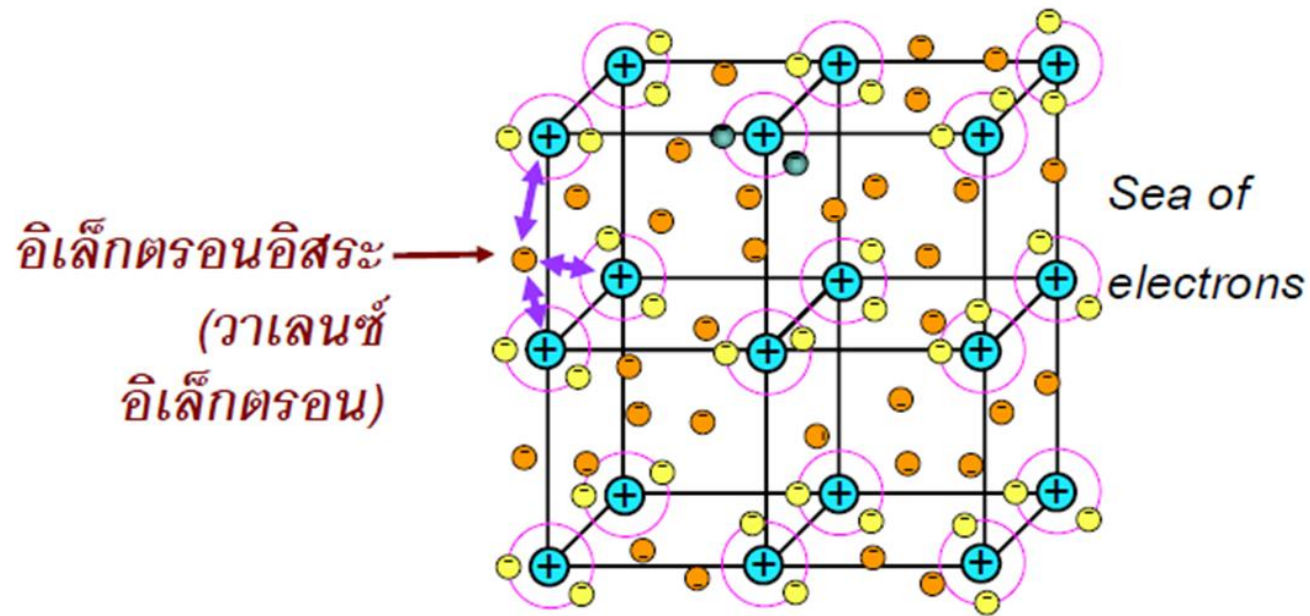
$$\approx -1016.6$$



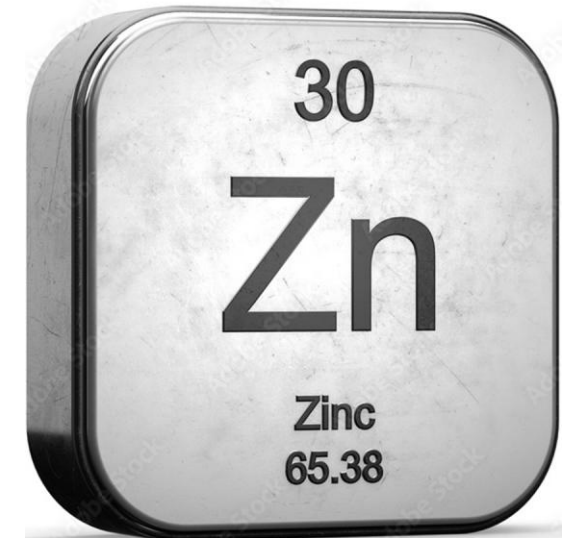
Metallic Bond

พันธะโลหะ คือแรงที่ยึดอะตอมโลหะไว้ด้วยกันในผลึก (ระหว่างอะตอมที่มีค่า EN ต่ำ)

- วาเลนซ์อิเล็กตรอนของโลหะเคลื่อนที่ไปในที่ต่าง ๆ ได้อย่างอิสระ
- อิเล็กตรอนอิสระทำหน้าที่ดึงดูดนิวเคลียสของอะตอมต่าง ๆ เข้าด้วยกัน

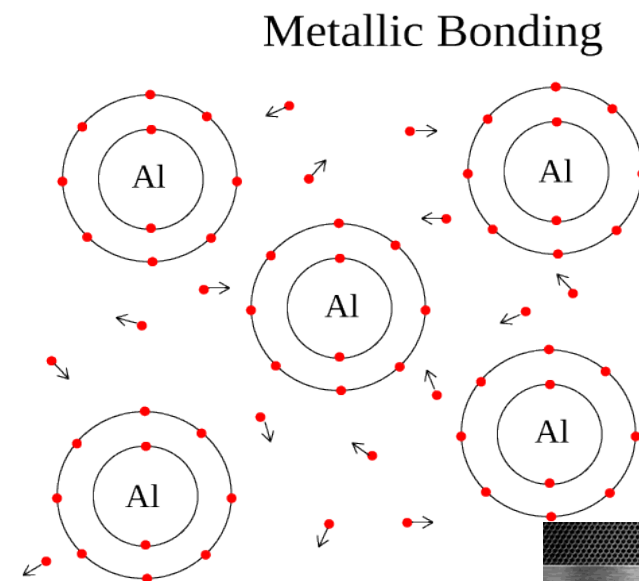
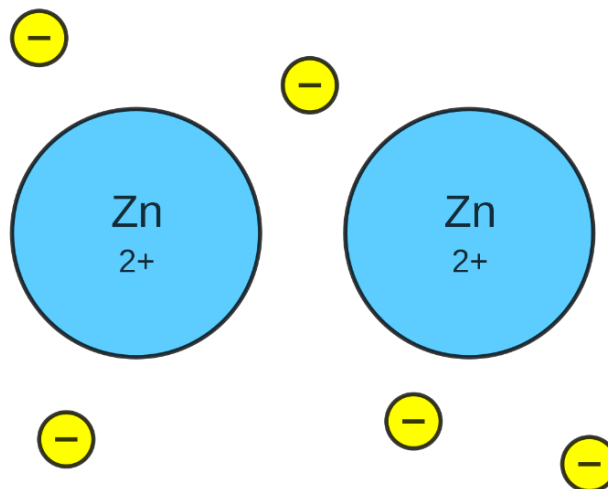
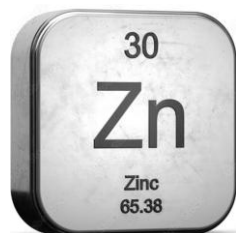
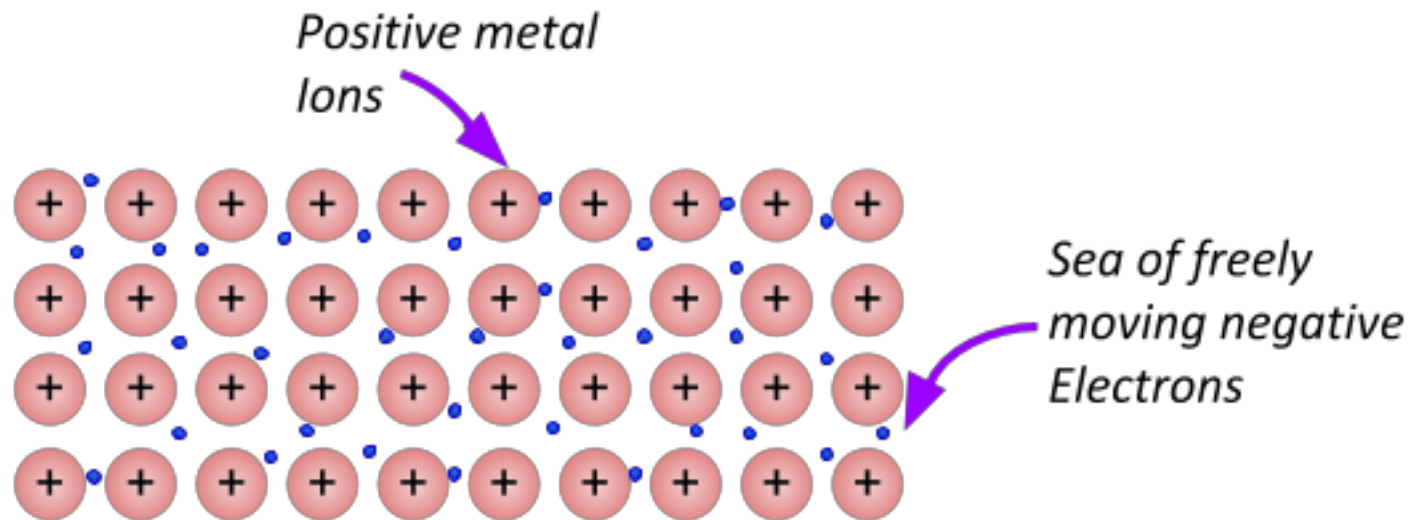
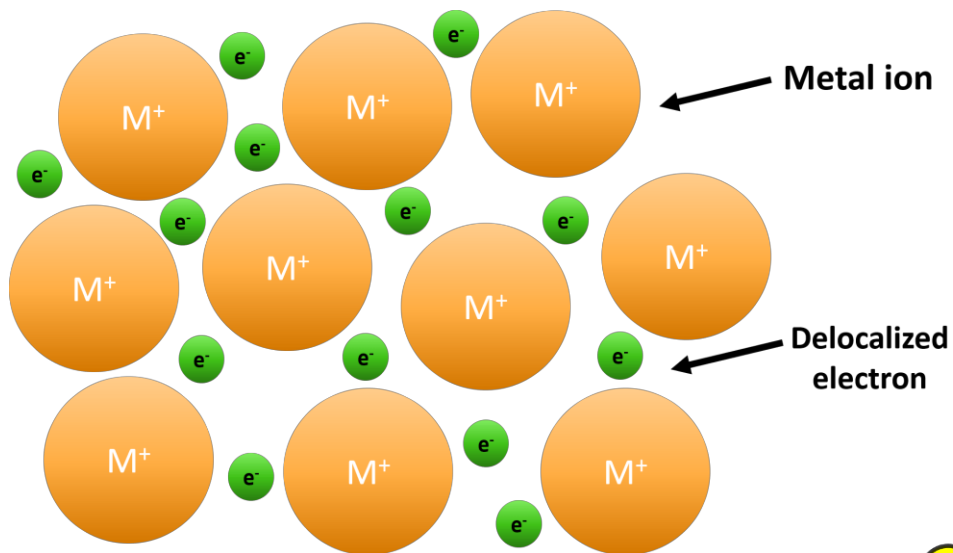


วาเลนซ์อิเล็กตรอนมาก → พันธะแข็งแรงมาก



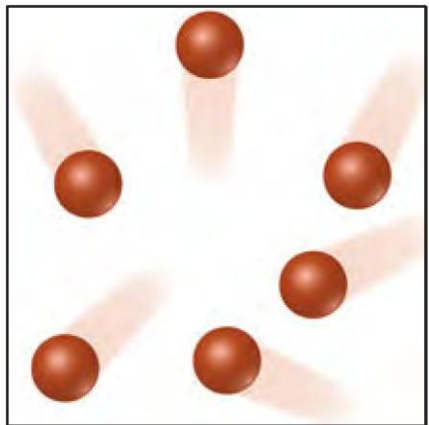


Metallic Bond

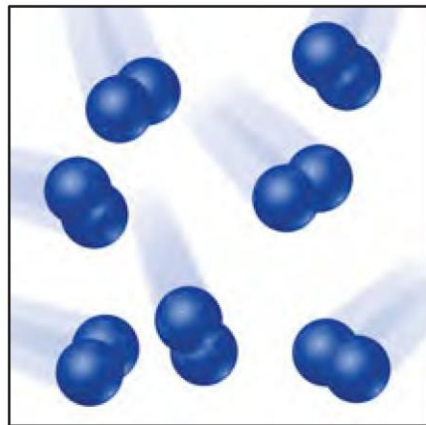




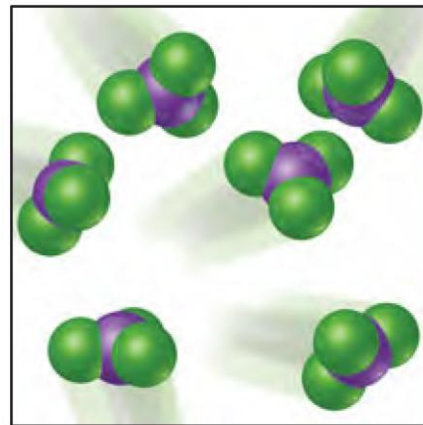
Naming Compounds: การเรียกชื่อสารประกอบ



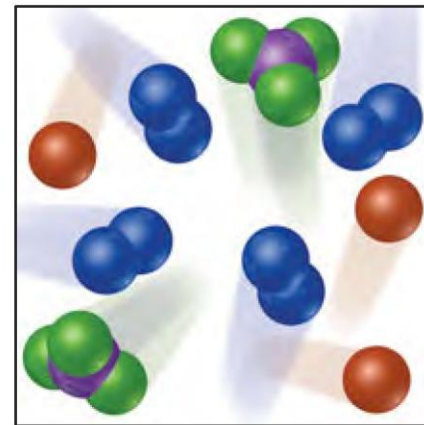
(a) Atoms of an element



(b) Molecules of an element



(c) Molecules of a compound



(d) Mixture of elements and a compound

Only one kind of atom is in any element.

Compounds must have at least two kinds of atoms.

The elemental composition of a compound is always the same that is known as the **law of constant composition** (or the law of definite proportions).



Naming Compounds

There are two broad classes of **binary compounds**.

1. Compounds that contain a metal and a nonmetal

(Types I and II)

2. Compounds that contain two non-metals

(Type III)

****Naming compounds can be divided into three types



Type I Binary Ionic Compounds

Rules for Naming **Type I** Ionic Compounds

1. The **cation** is always named first and the anion second.

2. A simple cation (obtained from a single atom) takes its name from the name of the element.

For example, Na^+ is called “sodium” in the names of compounds containing this ion.

3. A simple anion (obtained from a single atom) is named by taking the first part of the element name (the root) and adding “-ide”.

For example, Cl^- ion is called “chloride.”



Type I Binary Ionic Compounds

Compound	Ions Present	Name
NaCl	Na^+ , Cl^-	sodium chloride
KI	K^+ , I^-	potassium iodide
CaS	Ca^{2+} , S^{2-}	calcium sulfide
CsBr	Cs^+ , Br^-	cesium bromide
MgO	Mg^{2+} , O^{2-}	magnesium oxide

Cation	Name	Anion	Name*
H^+	hydrogen	H^-	hydride
Li^+	lithium	F^-	fluoride
Na^+	sodium	Cl^-	chloride
K^+	potassium	Br^-	bromide
Cs^+	cesium	I^-	iodide
Be^{2+}	beryllium	O^{2-}	oxide
Mg^{2+}	magnesium	S^{2-}	sulfide
Ca^{2+}	calcium		
Ba^{2+}	barium		
Al^{3+}	aluminum		
Ag^+	silver		
Zn^{2+}	zinc		

Note: There are Mg(I) and Ca(I) compounds had been reported recently in literature; however, they are not stable under normal atmosphere.



Type II Binary Ionic Compounds

There are many metals that can form more than one type of cations. For example, lead (Pb) can form Pb^{2+} or Pb^{4+} in ionic compounds.

- Iron (Fe) ---- Fe^{2+} , Fe^{3+}
- Chromium (Cr) --- Cr^{2+} , Cr^{3+}
- Thus, if we say “iron chloride” it could mean FeCl_2 and FeCl_3 .
- Chemists have decided to deal with this situation by using a Roman numeral to specify the charge on the cation.
- FeCl_2 = iron(II) chloride
- FeCl_3 = iron(III) chloride



Type II Binary Ionic Compounds

Ion	Systematic Name	Older Name
Fe^{3+}	iron(III)	ferric
Fe^{2+}	iron(II)	ferrous
Cu^{2+}	copper(II)	cupric
Cu^{+}	copper(I)	cuprous
Co^{3+}	cobalt(III)	cobaltic
Co^{2+}	cobalt(II)	cobaltous
Sn^{4+}	tin(IV)	stannic
Sn^{2+}	tin(II)	stannous
Pb^{4+}	lead(IV)	plumbic
Pb^{2+}	lead(II)	plumbous
Hg^{2+}	mercury(II)	mercuric
Hg_2^{2+*}	mercury(I)	mercurous

Anion	Name*
H^{-}	hydride
F^{-}	fluoride
Cl^{-}	chloride
Br^{-}	bromide
I^{-}	iodide
O^{2-}	oxide
S^{2-}	sulfide



Type III Binary Compounds

Naming Binary Compounds that Contain only Nonmetals (Type III)

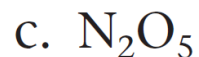
Rules of Naming Type III Binary Compounds

1. The first element in the formula is named first, and the full element name is used.
2. The second element is named as though it were an anion.
3. Prefixes are used to denote the numbers of atoms present.
4. The prefix mono- is never used for naming the first element.
For example, CO is called “carbon monoxide, not monocarbon monoxide.”



Type III Binary Compounds

Name the following binary compounds, which contain two nonmetals (Type III).

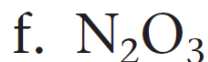
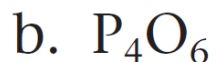


Boron tri fluoride

Nitrogen monoxide

dinitrogen
pentaoxide

Name each of the following compounds.



phosphorus pentachloride

Sulphur hexafluoride

สาร	ชื่อ
CO	Carbon monoxide
CO_2	Carbon dioxide
BF_3	Boron trifluoride
SiCl_4	Silicon tetrachloride
SF_6	Sulphur hexafluoride
P_2O_5	Diphosphorus pentoxide
P_4O_{10}	Tetraphosphorus decaoxide
Cl_2O_7	Dichlorine heptoxide

Table 5.3 Prefixes Used to Indicate Numbers in Chemical Names

Prefix	Number Indicated
<i>mono-</i>	1
<i>di-</i>	2
<i>tri-</i>	3
<i>tetra-</i>	4
<i>penta-</i>	5
<i>hexa-</i>	6
<i>hepta-</i>	7
<i>octa-</i>	8



Valence bond theory (VB)

Ref: Ralph H. Petrucci, F. Geoffrey Herring, Jeffry D. Madura, Carey Bissonnette. General Chemistry: Principles and Modern Applications. Prentice Hall 2010, English 10th ed.

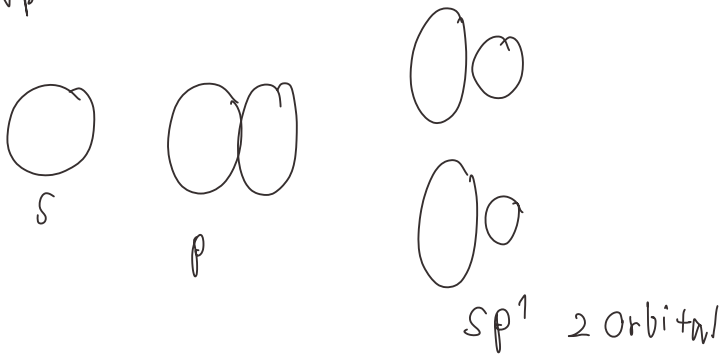
Theodore E. Brown, H. Eugene H LeMay, Bruce E. Bursten, Catherine Murphy, Patrick Woodward. Chemistry: The Central Science. Prentice Hall 2011, English 12th ed.

Raymond Chang and Jason Overby. General Chemistry: The Essential Concepts. McGraw-Hill Science/Engineering/Math 2010, 6th ed.

Hybridization = n. พหุ orbital ผสมกัน

• จพ. Hybrid orbital = จพ. orbital ตั้งขึ้นก่อนหน้าผสม (เข้าทำใน 1 ออกทำ 1 น)

• sp^1



1s orbital + 1p orbital
= 2 sp^1 hybrid orbital
เหลือ p ว่าง 2 orbital

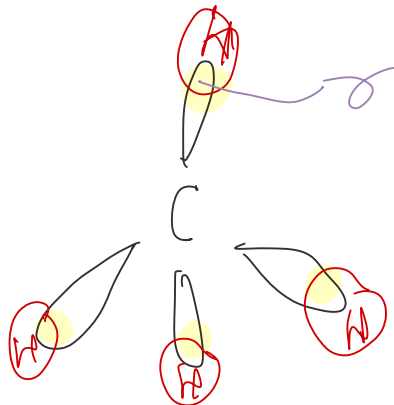
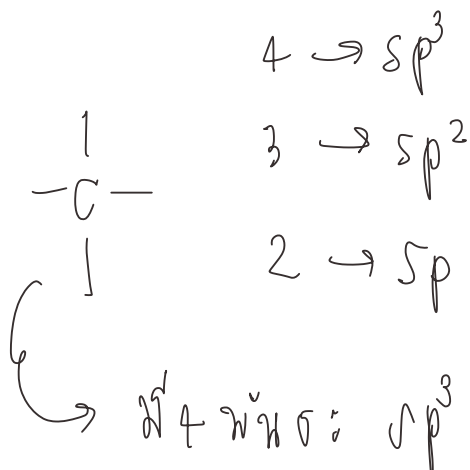
• sp^2

1s orbital + 2p orbital
= 3 sp^2 hybrid orbital 120°
เหลือ p ว่าง 1 orbital

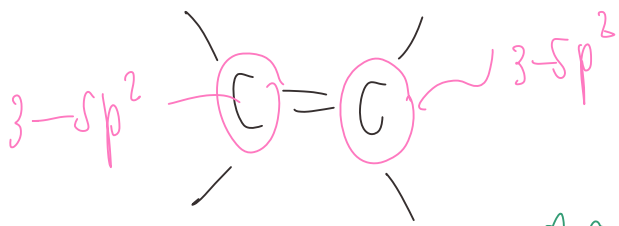
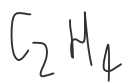
• sp^3

1s orbital + 3p orbital
= 4 sp^3 hybrid orbital ทรง 4 ขั้ว
เหลือ p ว่าง 0 orbital

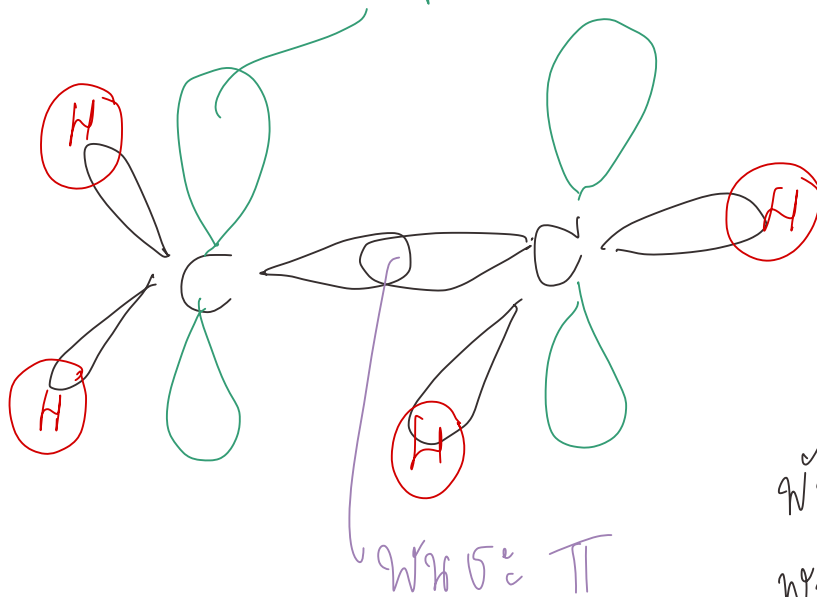
CH_4



σ 4 พันธะ



orbital ที่ใช้ 2 ตัว

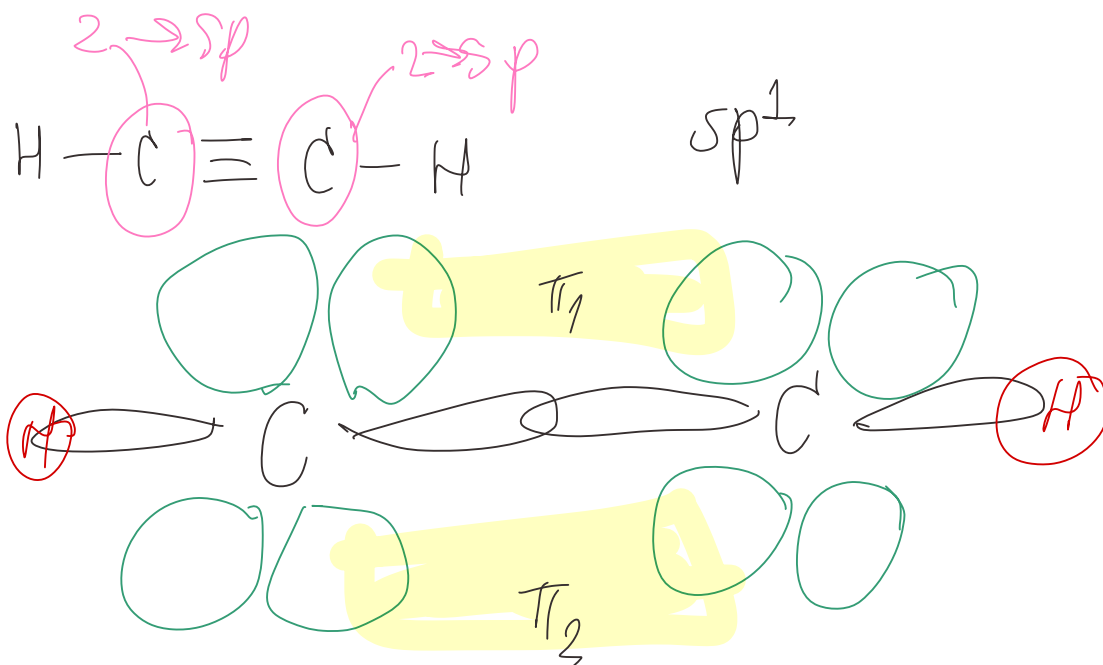


พันธะ π

พันธะเดี่ยว $\rightarrow \sigma$
พันธะคู่ $\rightarrow \sigma, \pi$

พันธะ π 1 พันธะ

พันธะ σ 5 พันธะ



3 σ -bond มี 3

2 π -bond มี 2

พันธะเดี่ยว $\rightarrow \sigma$

พันธะคู่ / สาม $\rightarrow \sigma + \pi$

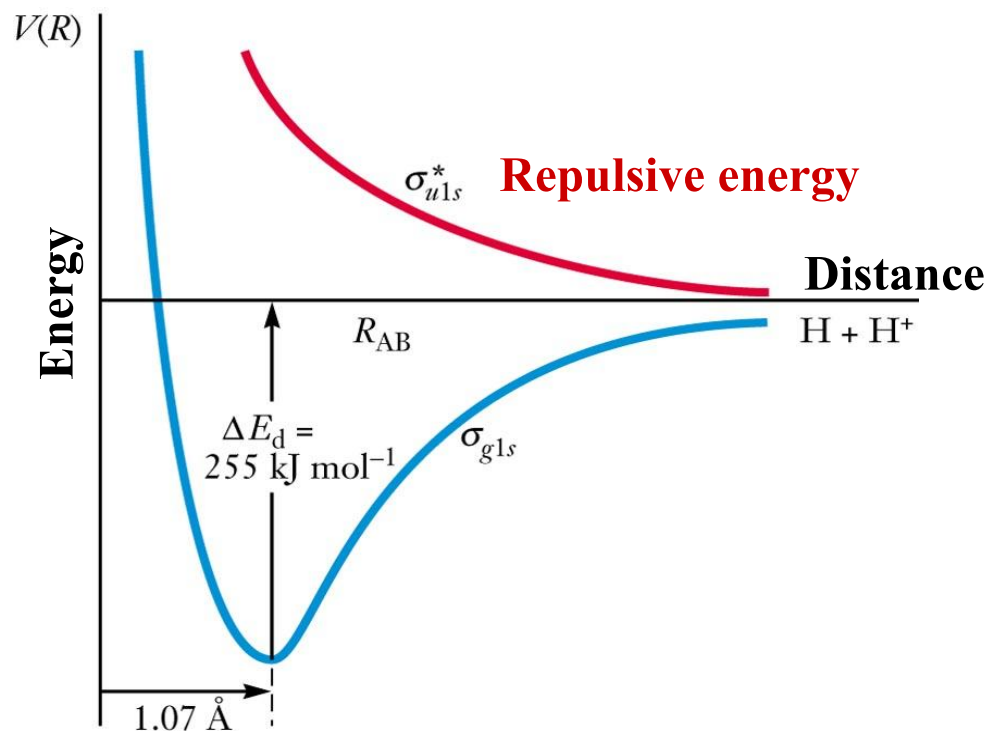
π

สาม = $\sigma + \pi$

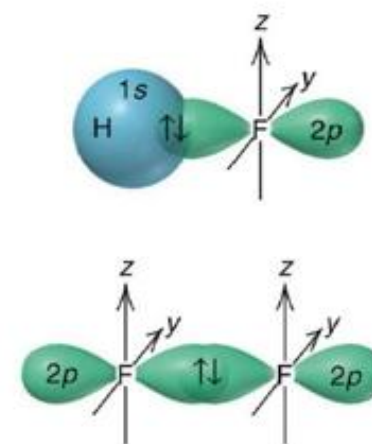
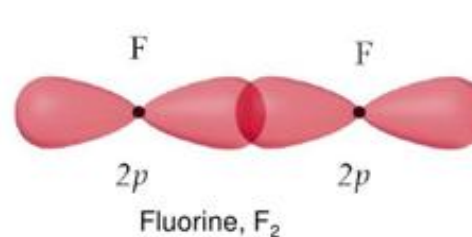
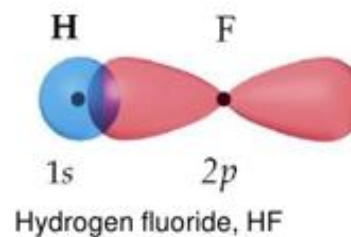
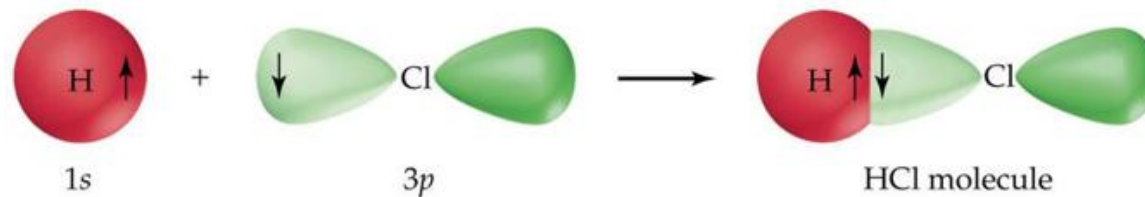


Valence bond theory (VB)

Valence bond (VB) theory assumes that all bonds are localized bonds formed between two atoms by the donation of an electron from each atom. This is actually an invalid assumption because many atoms bond using delocalized electrons. VB theory does a good job of qualitatively describing the shapes of covalent compounds.



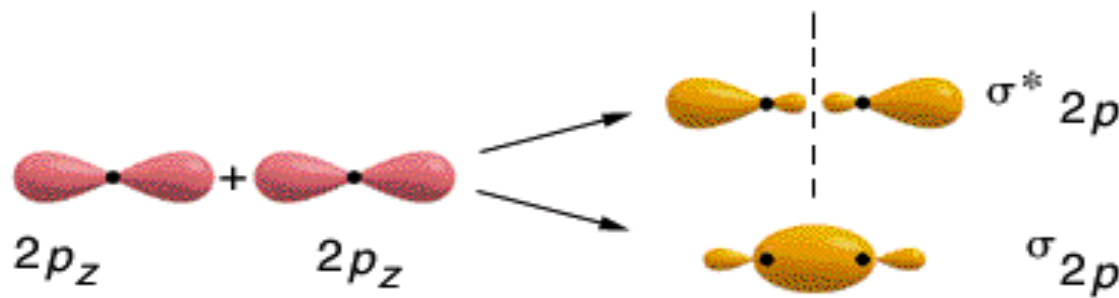
© 2003 Thomson-Brooks/Cole





Overlaps of p-p orbitals

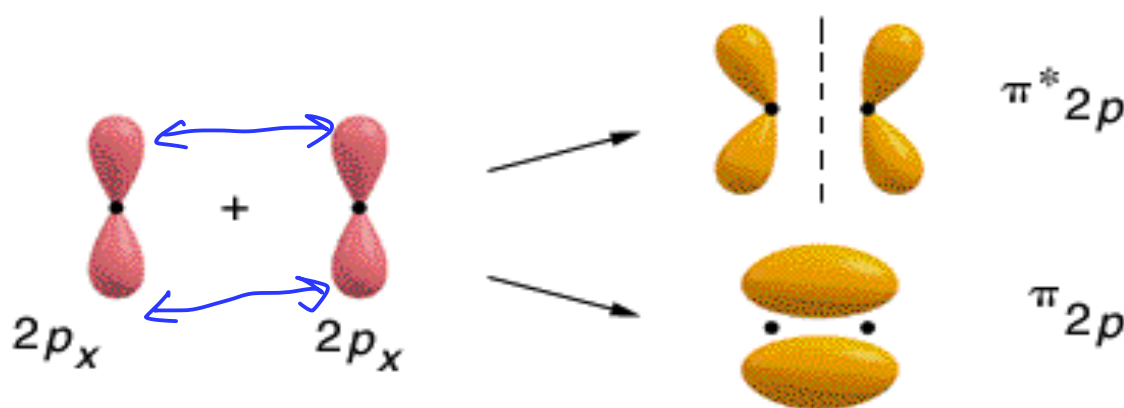
(a)



anti- σ -bond

σ -bond

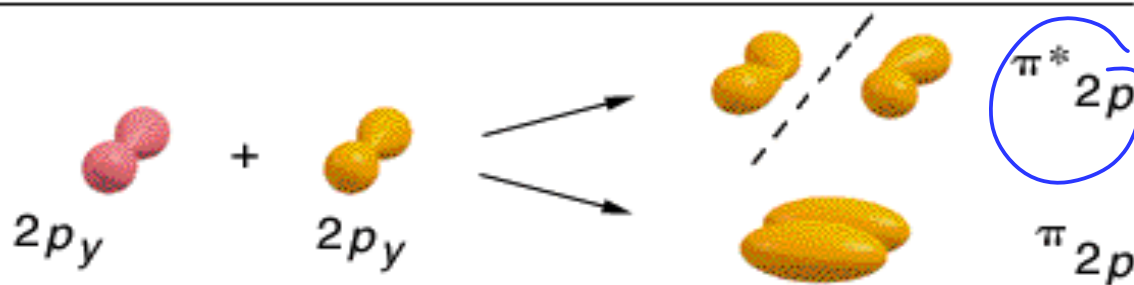
(b)



anti π -bond

π -bond

(c)



anti π bond

π -bond

z

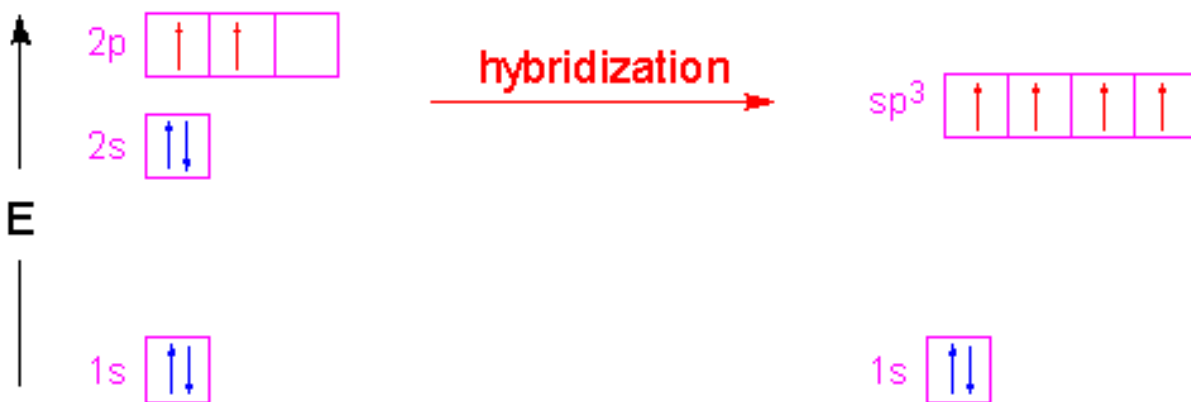
x

y

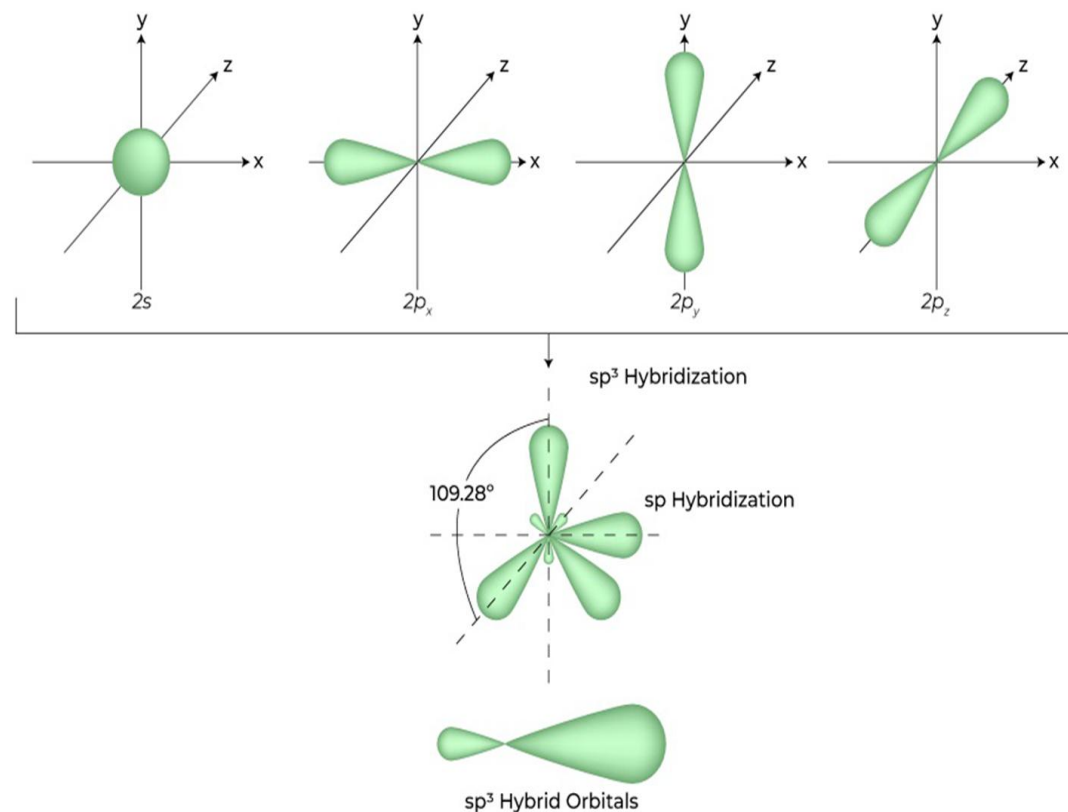


Hybridization

Hybridization is the idea that atomic orbitals fuse to form newly hybridized orbitals, which in turn, influences molecular geometry and bonding properties. Hybridization is also an expansion of the valence bond theory. In order to explore this idea further, we will utilize three types of hydrocarbon compounds to illustrate sp^3 , sp^2 , and sp hybridization.



sp^3 Hybridization

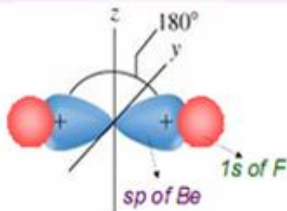
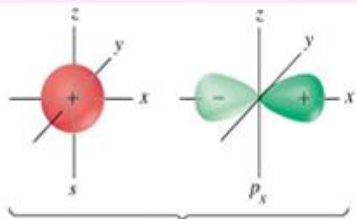
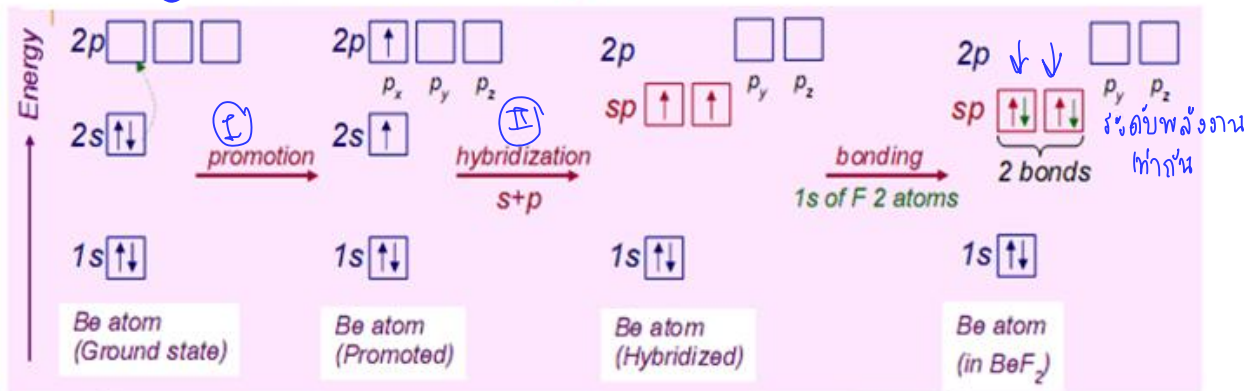




Hybridizations

1 sp Hybridization ใน BeF_2 $\text{Be} = 1s^2 2s^2$

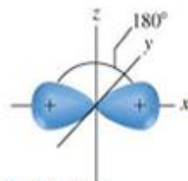
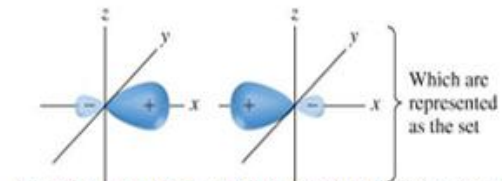
อะตอม Be



sp = 2 hybrid orbitals → 2 bonds

VSEPR = AX₂

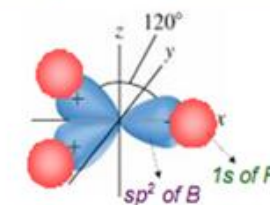
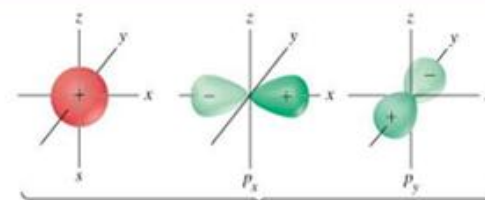
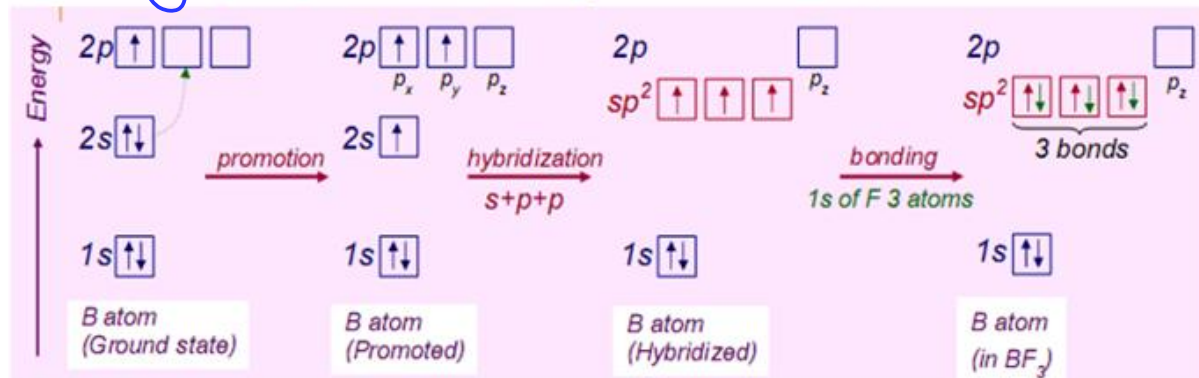
เส้นตรง



http://cwx.prenhall.com/petrucci/medialib/power_point/index.html



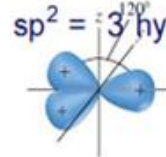
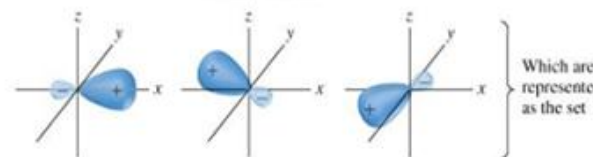
2 sp² Hybridization ใน BF_3 $\text{B} = 1s^2 2s^2 2p^1$



sp² = 3 hybrid orbitals → 3 bonds

VSEPR = AX₃

Δ ระนาบ





Hybridizations

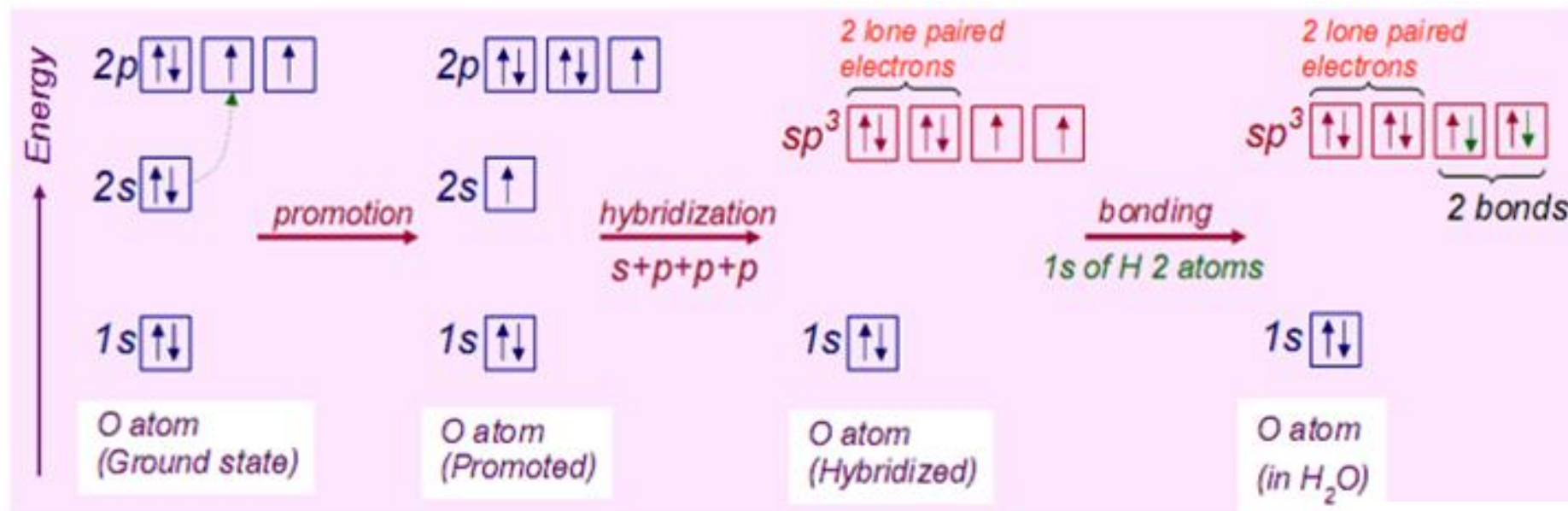
Regions of Electron Density	Arrangement		Hybridization	
2		linear	sp	
3		trigonal planar	sp^2	
4		tetrahedral	sp^3	
5		trigonal bipyramidal	sp^3d	
6		octahedral	sp^3d^2	

TABLE 9.4 Geometric Arrangements Characteristic of Hybrid Orbital Sets

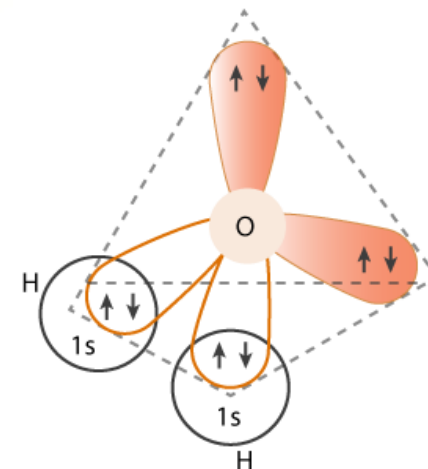
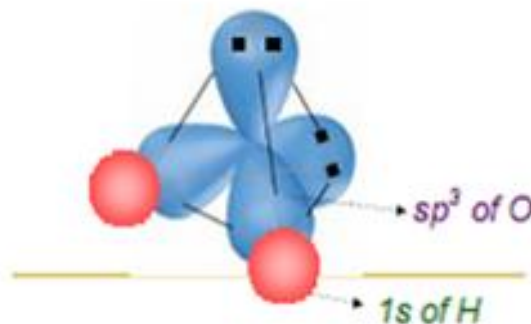
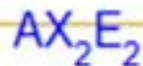
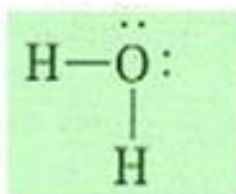
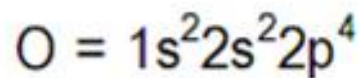
Atomic Orbital Set	Hybrid Orbital Set	Geometry	Examples
s, p	Two sp	 Linear	$\text{BeF}_2, \text{HgCl}_2$
s, p, p	Three sp^2	 Trigonal planar	BF_3, SO_3
s, p, p, p	Four sp^3	 Tetrahedral	$\text{CH}_4, \text{NH}_3, \text{H}_2\text{O}, \text{NH}_4^+$
s, p, p, p, d	Five sp^3d	 Trigonal bipyramidal	$\text{PF}_5, \text{SF}_4, \text{BrF}_3$
s, p, p, p, d, d	Six sp^3d^2	 Octahedral	$\text{SF}_6, \text{ClF}_5, \text{XeF}_4, \text{PF}_6^-$



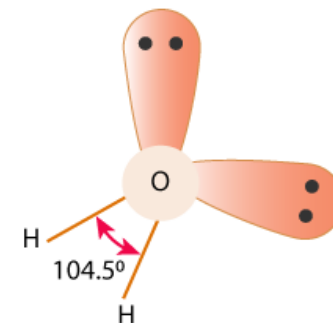
Hybridization of H₂O



sp^3 hybridization



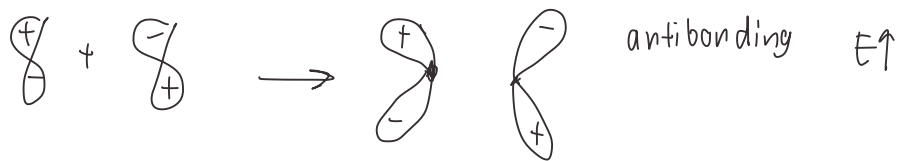
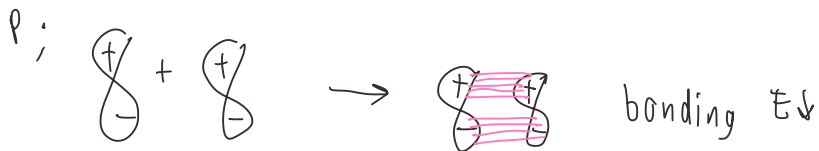
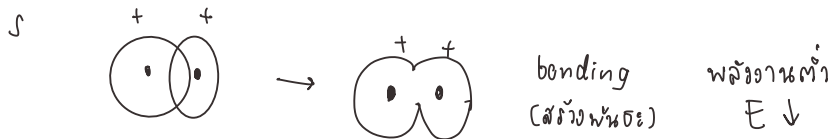
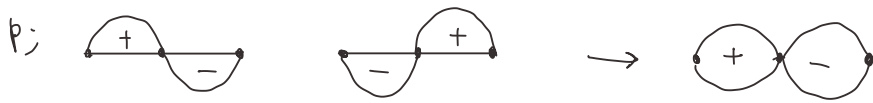
Orbital overlap in H₂O



Bent shape

molecular orbital theory

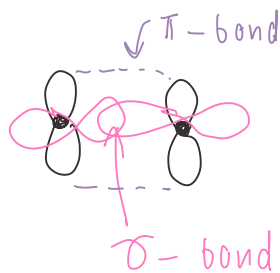
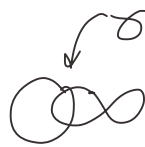
orbital; s p d *



σ, π

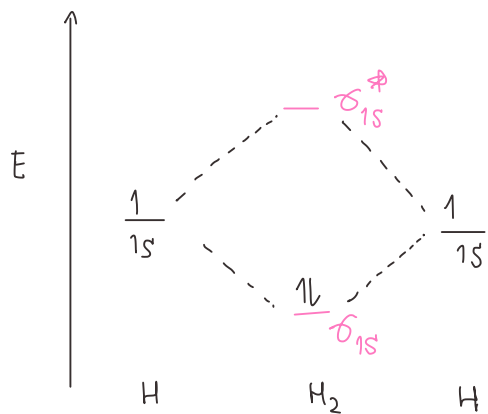
σ = sigma (หัวชนหัว)

π = pi (ข้างๆ ไขว้กัน)



H₂ ; 1s¹

bonding กับ antibonding



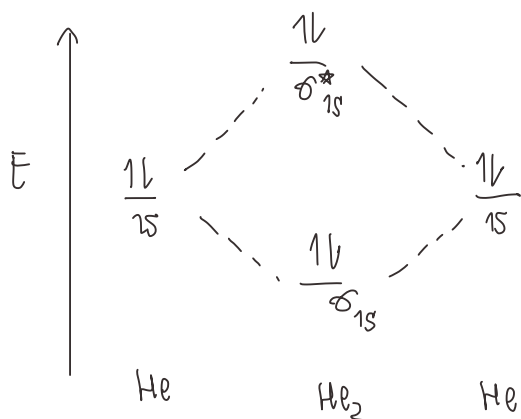
Bonding Order : $\frac{1}{2}$ (bonding - antibonding)
BO

$$\therefore BO = \frac{1}{2} (2 - 0) = 1$$

Diamagnetic

↑
พันธะเดี่ยว

He₂ ; He : 1s²

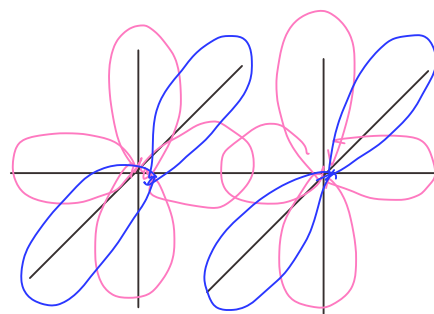
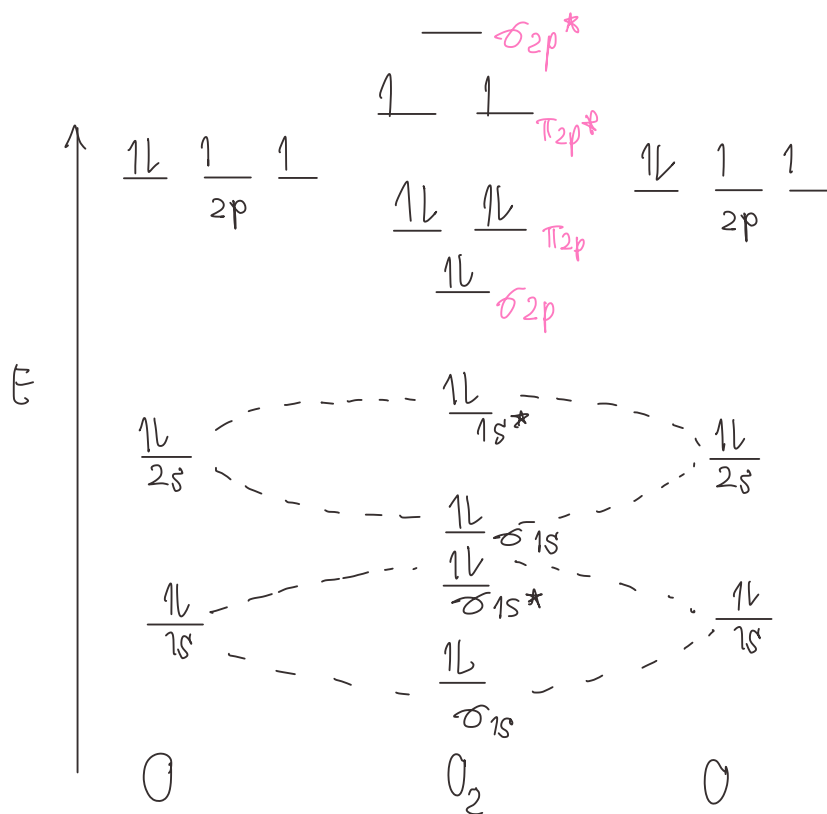


$$B.O. = \frac{1}{2} (2 - 2) = 0$$

diamagnetic

↑
ไม่ได้สร้างพันธะกันเลย

O₂ ; 6O = 1s² 2s² 2p⁴



$$B.O. = \frac{1}{2} (10 - 6) = \frac{4}{2} = 2$$

paramagnetic

↑
พหุคูณ



Molecular Orbitals (MO)

Ref: Ralph H. Petrucci, F. Geoffrey Herring, Jeffry D. Madura, Carey Bissonnette. General Chemistry: Principles and Modern Applications. Prentice Hall 2010, English 10th ed.

Theodore E. Brown, H. Eugene H LeMay, Bruce E. Bursten, Catherine Murphy, Patrick Woodward. Chemistry: The Central Science. Prentice Hall 2011, English 12th ed.

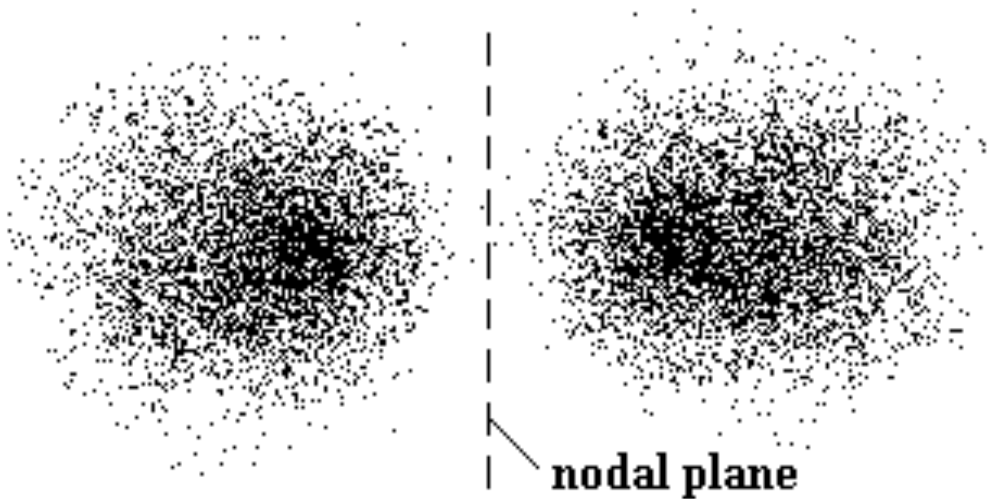
Raymond Chang and Jason Overby. General Chemistry: The Essential Concepts. McGraw-Hill Science/Engineering/Math 2010, 6th ed.



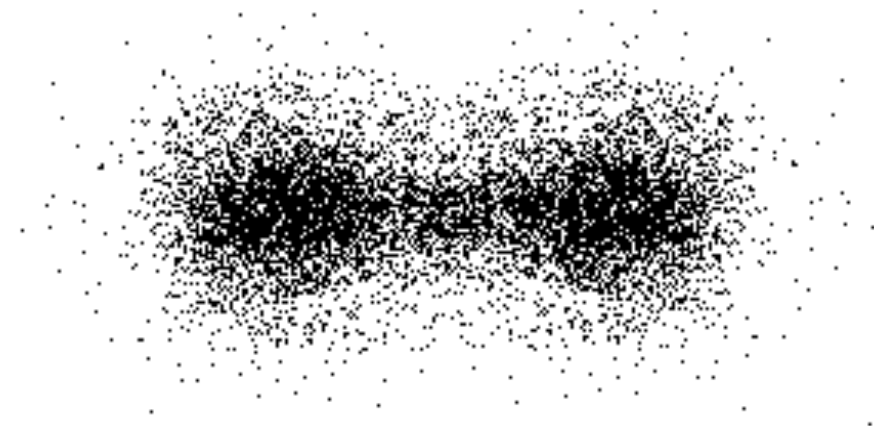
Formation of Molecular Orbitals (MO) from Atomic Orbitals

- Schrodinger equations can be written for electrons in molecules.
- Approximate solutions to these molecular Schrodinger equations can be constructed from **linear combinations of atomic orbitals (LCAO)**, the sums and differences of the atomic wave functions.

$$\Psi = c_a\psi_a + c_b\psi_b$$



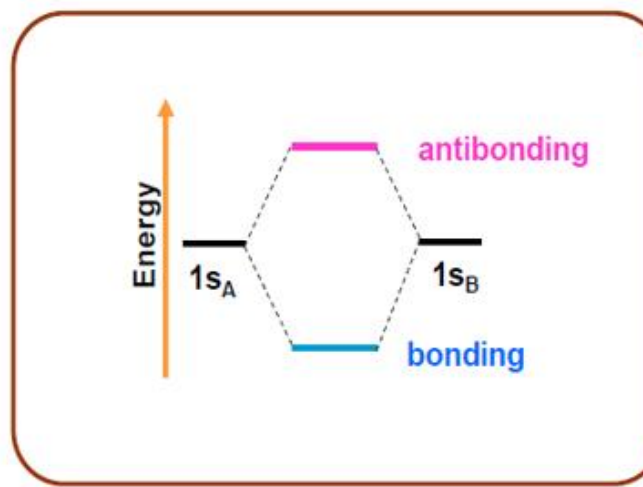
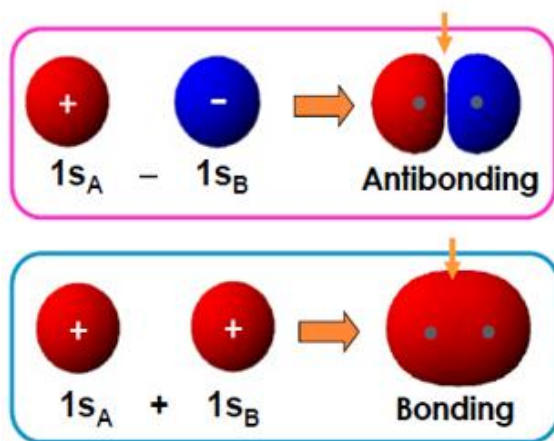
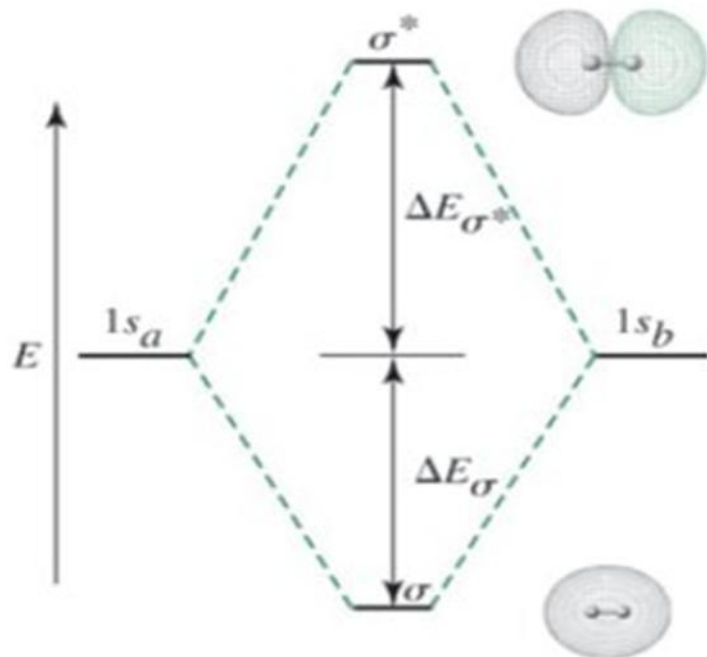
Anti-Bonding



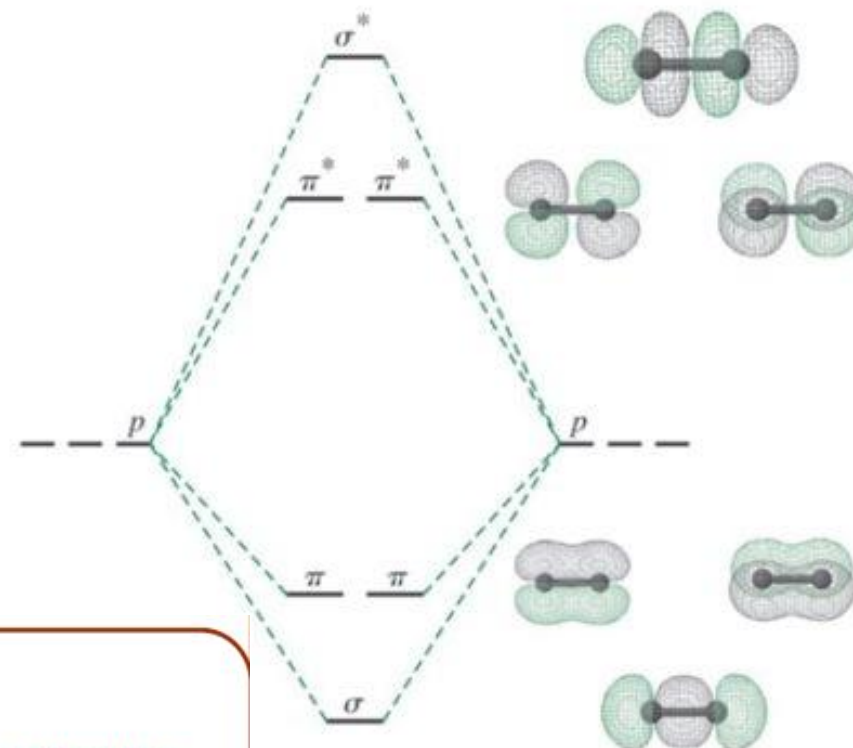
Bonding



Bonding of s-Orbitals



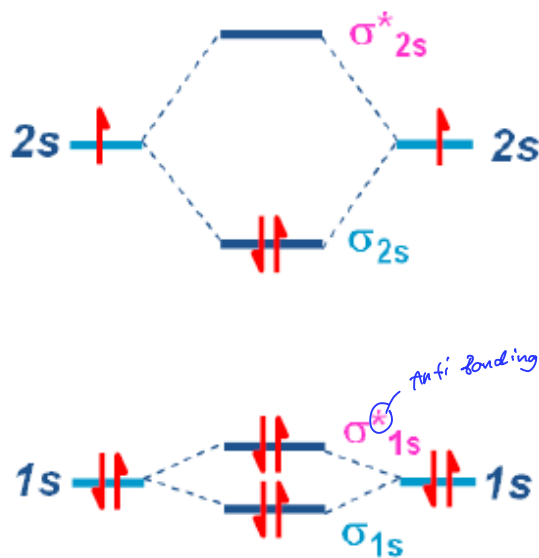
Bonding of p-Orbitals



MO Diagram of Li_2 & Be_2

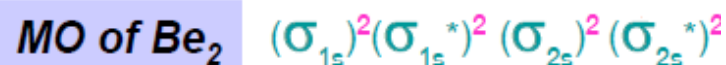
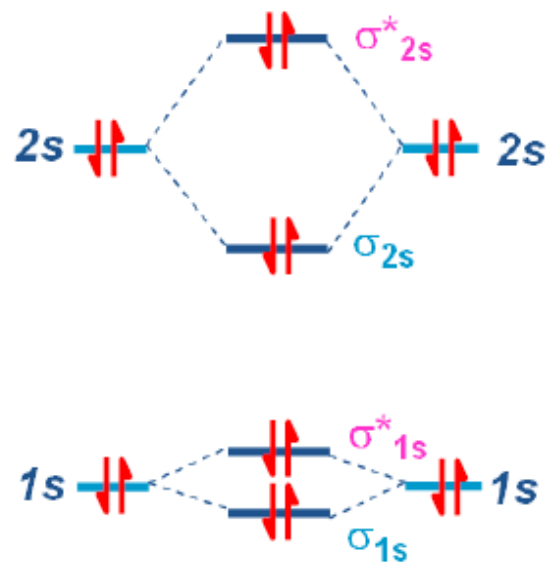
- Li_2 (2 x 3 electrons)

- $\text{Li} = 1s^2 2s^1$
- Bond order = 1



- Be_2 (2 x 4 electrons)

- $\text{Be} = 1s^2 2s^2$
- Bond order = 0 (no bond)

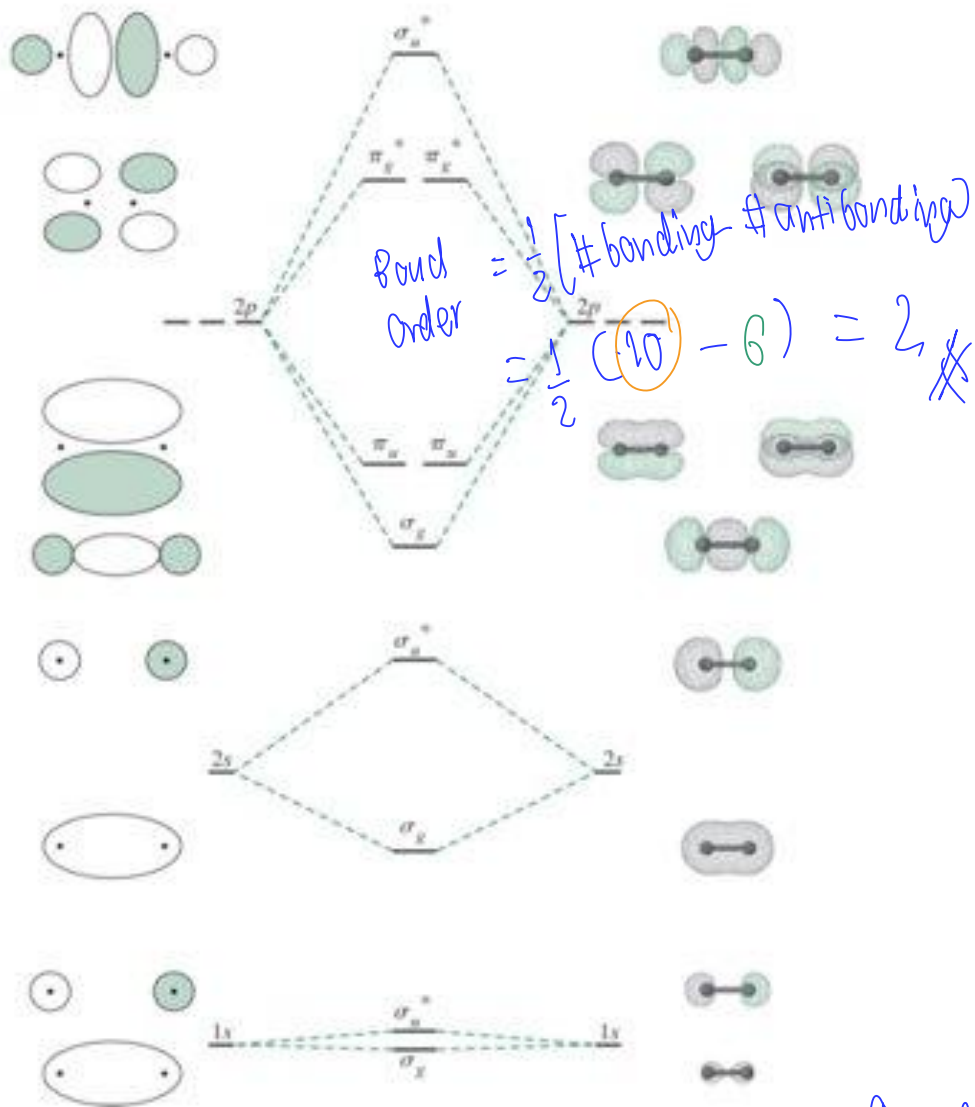


$$\text{Bond order} = \frac{1}{2} \left[\left(\text{number of electrons in bonding orbitals} \right) - \left(\text{number of electrons in antibonding orbitals} \right) \right]$$

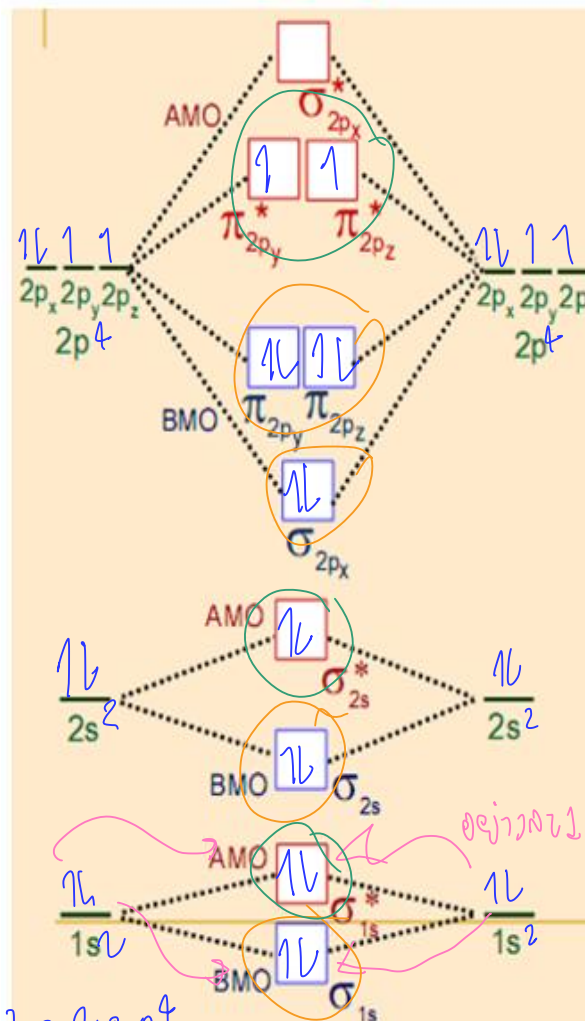


Homonuclear Diatomic Molecules

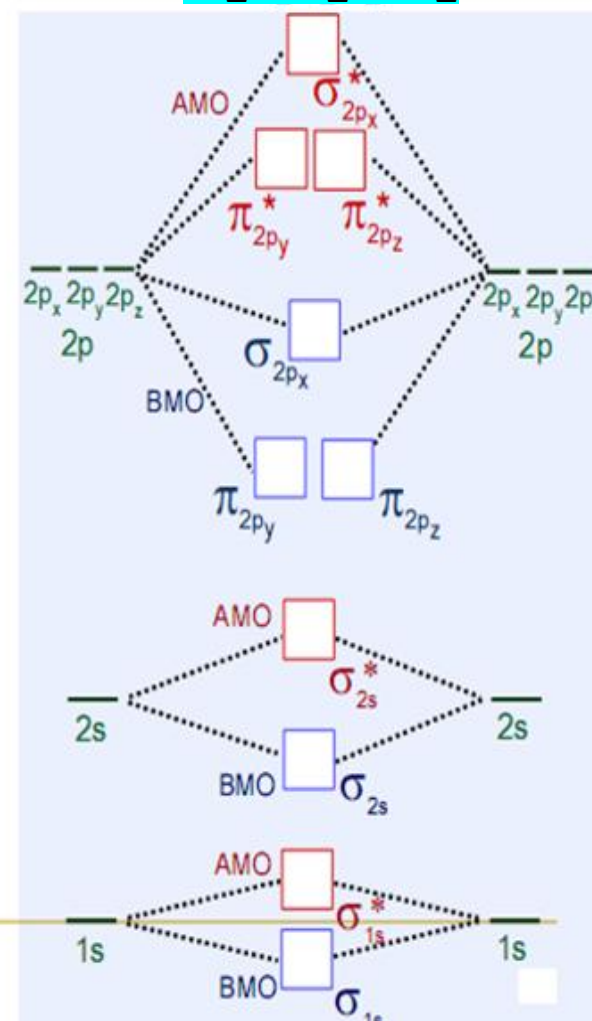
MO diagrams



O₂, F₂, Ne₂



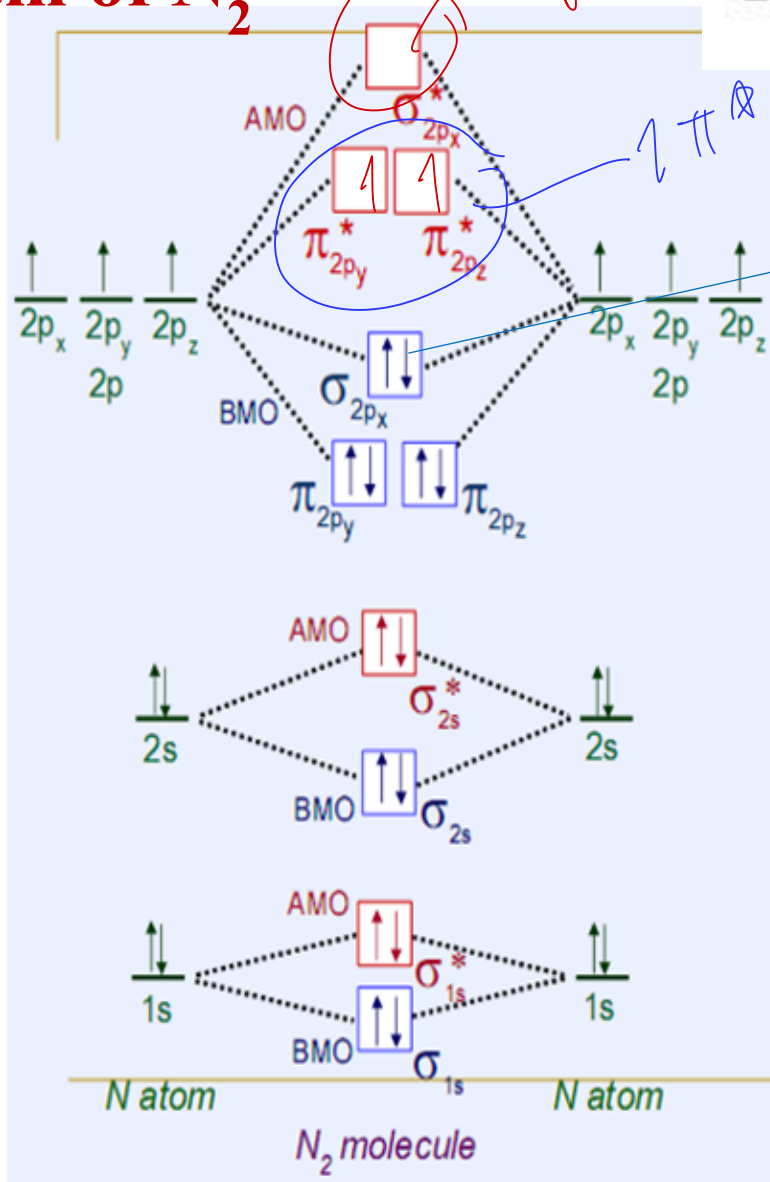
B₂, C₂, N₂



$\sigma 1s = 1s^2 2s^2 2p^4$



Ex. MO diagram of N₂



$$\text{Bond order} = \frac{1}{2} \left[\left(\text{number of electrons in bonding orbitals} \right) - \left(\text{number of electrons in antibonding orbitals} \right) \right]$$

Diamagnetic

N₂ =

$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2$$

$$(\pi_{2py})^2 (\pi_{2pz})^2 (\sigma_{2px})^2$$

อันดับพันธะของ N₂ = $\frac{1}{2} (10 - 4) = 3$

Bond order = 3

1 σ bond
2 π bond

1 σ bond

2 π bond

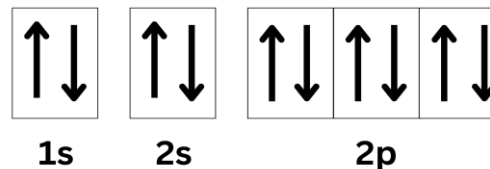


Types of magnetism

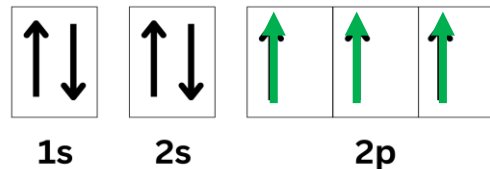
Spin alignment with
magnetic field **B**



Diamagnetic (paired electrons)



Paramagnetic (unpaired electrons)



Paramagnetic

s orbital *p orbitals*

- Unpaired electrons
- Attracted to a magnetic field

Diamagnetic

s orbital *p orbitals*

- Paired electrons
- Repelled by a magnetic field

He

Na
Sodium

Al

Pt

U

He

C

Cu

Ag

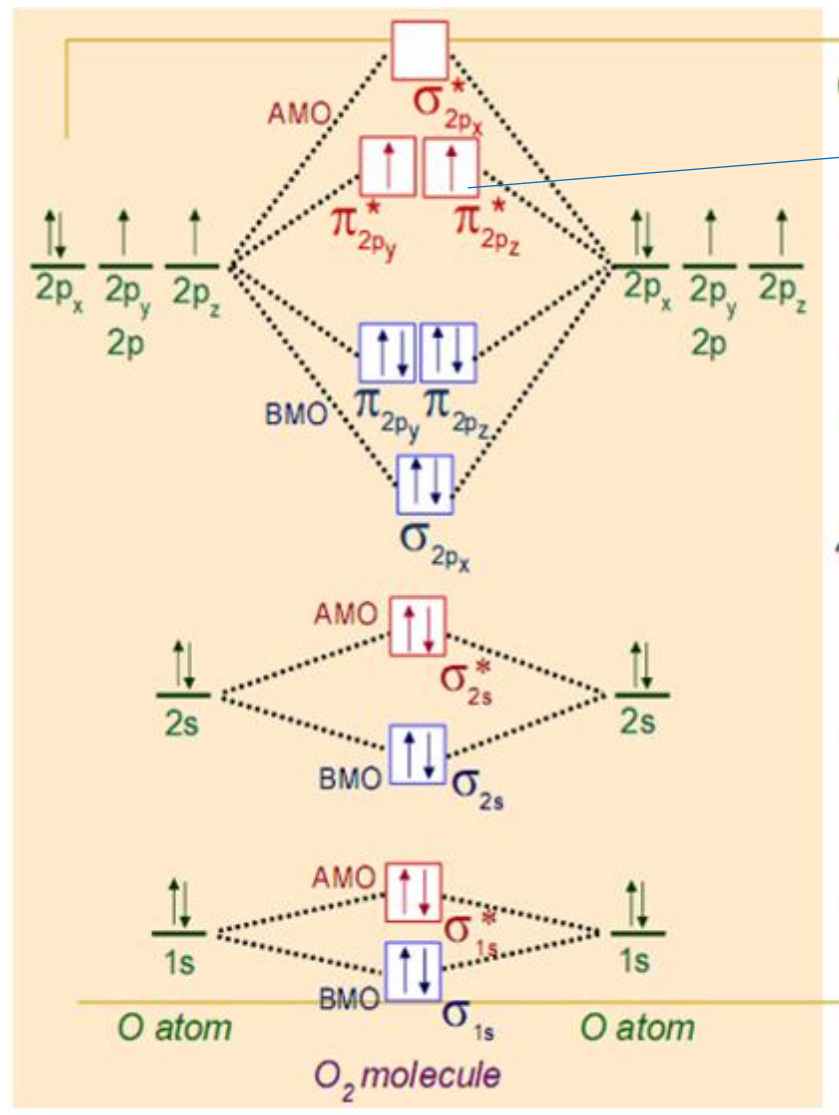
Au

All materials are diamagnetic, but in atoms with unpaired electrons, paramagnetism overcomes diamagnetism.

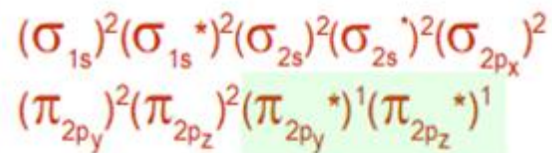
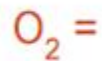
sciencenotes.org



Ex. MO diagram of O₂



→ Paramagnetic



อันดับพันธะของ O₂ = $\frac{1}{2} (8-4) = 2$

1 σ bond
1 π bond

Bond order = 2

$$\text{Bond order} = \frac{1}{2} \left[\left(\text{number of electrons in bonding orbitals} \right) - \left(\text{number of electrons in antibonding orbitals} \right) \right]$$



MO of Homonuclear diatomic molecules

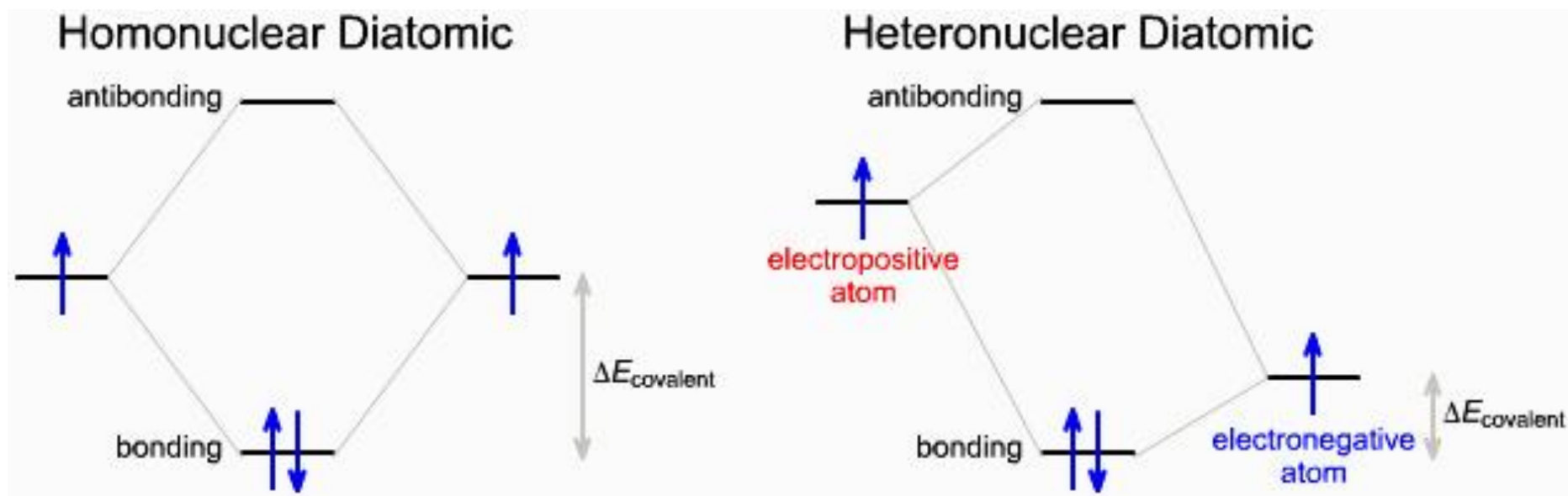
No 2s-2p interaction		Small 2s-2p interaction			Large 2s-2p interaction		
B ₂	Be ₂	B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
Bond order = 1 Diamagnetic	Bond order = 0 Unstable	Bond order = 1 Paramagnetic	Bond order = 2 Diamagnetic	Bond order = 3 Diamagnetic	Bond order = 2 Paramagnetic	Bond order = 1 Diamagnetic	Bond order = 0 Unstable



MO of Heteronuclear Diatomic Molecule

HF Hydrogen fluoride

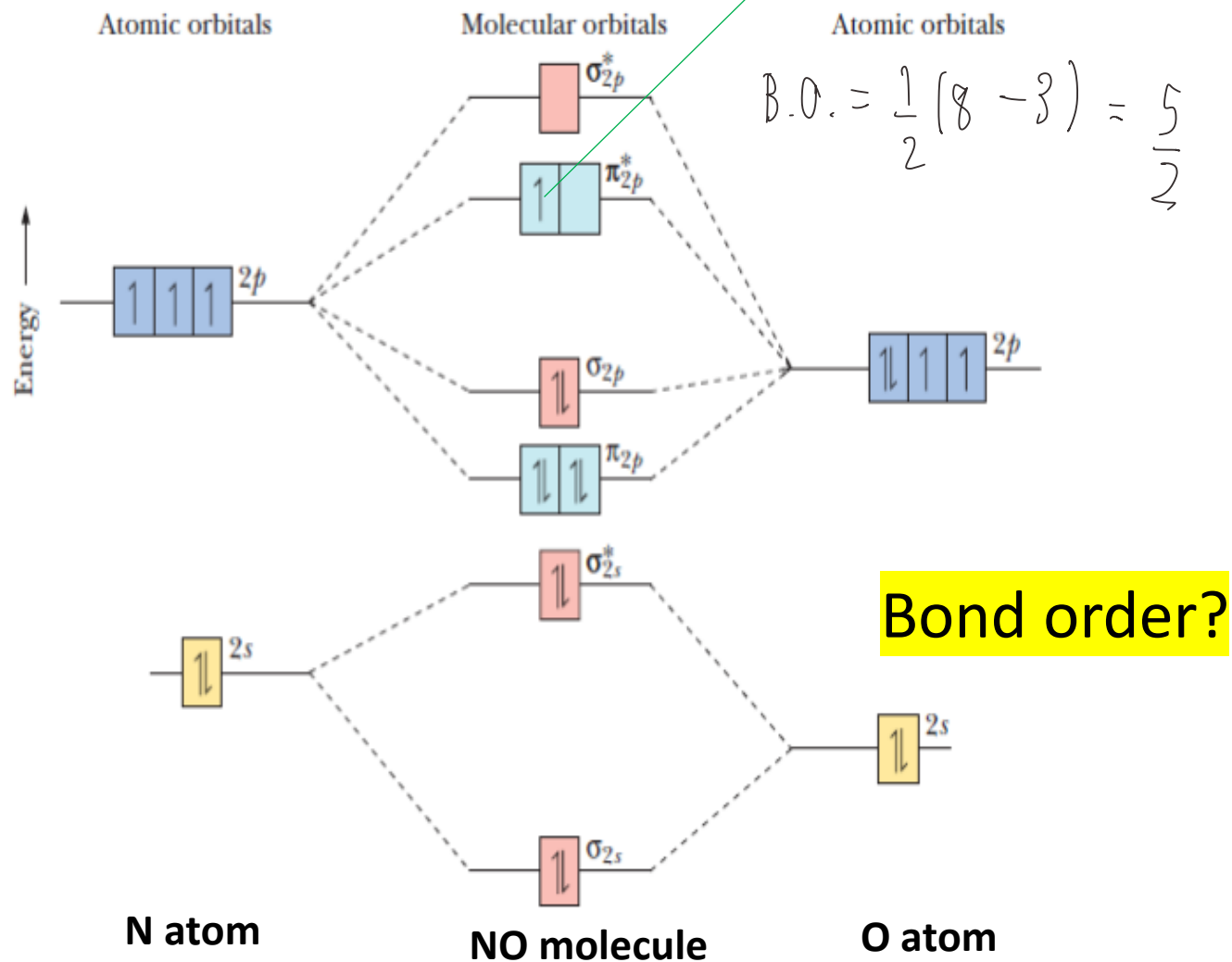
CO Carbon monoxide





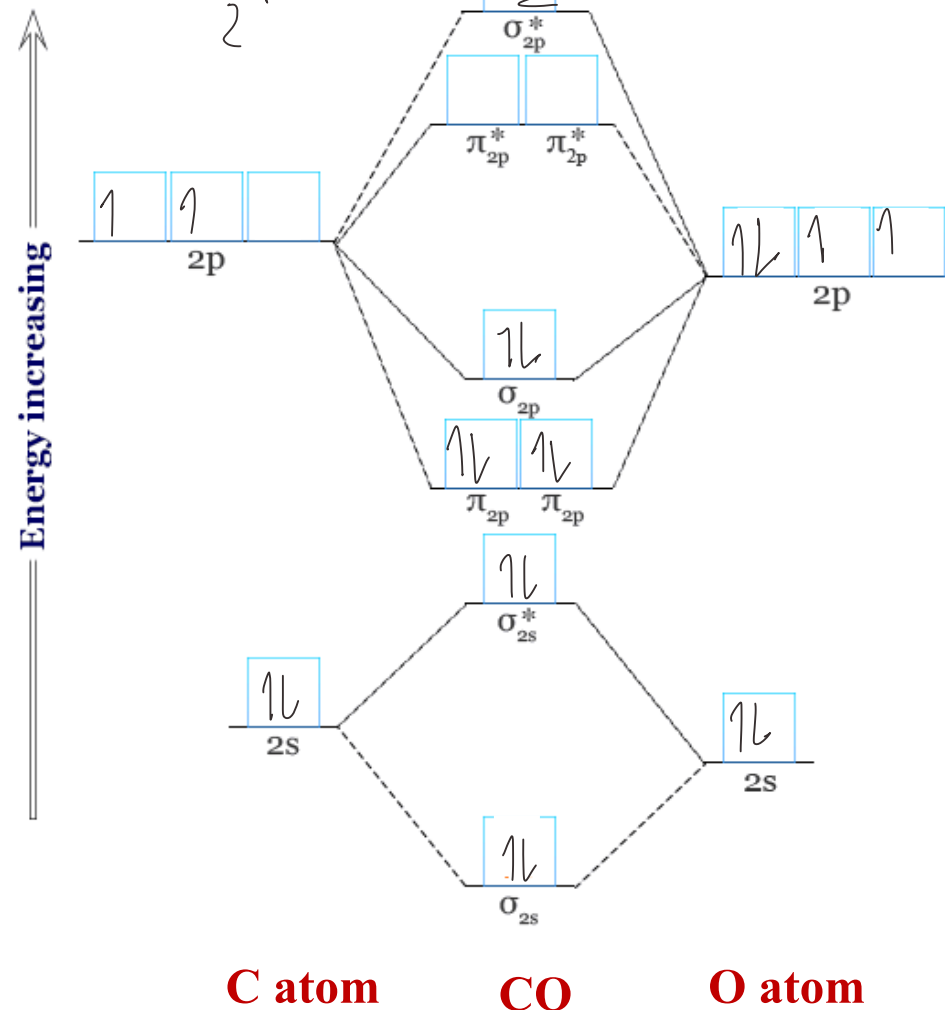
Ex. MO diagram of NO

Paramagnetic



Ex. Write MO diagram of CO

$$\text{B.O.} = \frac{1}{2}(8 - 2) = \frac{6}{2} = 3$$





See you next time!



**Our Soul is for the
Benefit of Mankind**

Ends